

Bachelor Thesis

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From Security by Obscurity to Imminent Exposure:

Security Risks for IoT Devices in the IPv6
Transition

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Abstract

The transition from IPv4 to IPv6 changes how residential Internet of Things (IoT) devices are exposed. Under IPv4, NAT-based residential deployments commonly discarded unsolicited inbound traffic by default, giving devices an implicit perimeter and limiting their direct reachability from the public Internet. IPv6 removes this form of implicit protection. Devices acquire globally routable addresses and, depending on the router configuration, may become directly reachable from the public Internet without any change to their own security properties.

Prior work has studied this shift only in fragments, including residential firewall behavior, IPv6 host discovery, device identifiability, and firmware weaknesses. The path from a globally addressed device to externally accessible services has received comparatively limited attention under controlled residential conditions. This thesis addresses that gap by deploying an ESP32 device running custom firmware on the lwIP stack and exposing deliberately weak HTTP and Telnet services on a real residential IPv6 connection. The device is observed across five experimental scenarios.

The results show that exposure was governed primarily by the residential firewall rather than by changes in the device itself. With all device-side conditions held constant, disabling the perimeter moved the same host from externally inaccessible services to direct external access to its HTTP endpoint, hidden administrative interface, and Telnet service. Filtering, however, reduced accessibility but not observability. The device remained reachable through ICMPv6 and identifiable through its EUI-64-derived address even when all inbound TCP traffic was blocked. In contrast, the IPv4/NAT baseline kept the same services unreachable without port forwarding. These findings suggest that IPv6 does not inherently increase device insecurity, but it does alter the conditions under which vulnerable services become reachable. In the scenarios examined, exposure depended largely on network-level configuration choices rather than on actions taken by end users.

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1. Introduction

This thesis explores how global IPv6 addressing increases the attack surface of the Internet of Things (IoT) environment. The study combines a comprehensive literature review and an empirical assessment of default system configurations through controlled laboratory demonstrations, highlighting how the adoption of new network standards can potentially affect the security of millions of end IoT devices.

1.1 The IPv4 to IPv6 Transition is the result of an inherent limitation in the design of IPv4's address space, which only supports 2^{32} unique addresses. A boundary already exceeded by the estimated 18,8 billion IoT devices alone in 2024 [1]. IPv6 mitigates this limitation by implementing a 128-bit architecture, offering a virtually infinite address space. Despite its technical advantage, the global adoption has been gradual due to the current scale of IPv4-based infrastructures and the cost of migration. Therefore, technologies such as Network Address Translation (NAT) were adopted to extend the lifetime of IPv4 [2].

In practice, NAT allows multiple internal hosts within a private local-area network (LAN) to share a single public IPv4 address by translating internal IP addresses and TCP/UDP port numbers to a public address and port at the network boundary. This translation process is stateful and connection-oriented, meaning that the address mappings are created only for outbound traffic initiated by hosts within the LAN [2], [3]. Consequently, unless explicitly allowed through static port-forwarding rules, any unsolicited inbound traffic that does not match an existing mapping entry is discarded by default [3]. Nonetheless, the use of NAT translation tables violates the original end-to-end connectivity principle of the Internet architecture [4], and simultaneously provides a form of perimeter protection by preventing direct external access to internal hosts.

1.2 Loss of NAT and Increased IoT Exposure reflects a shift in network architecture paradigms. Until now, many IoT devices have been designed under an implicit threat model that assumes limited inbound reachability, relying primarily on network topology, firewall configurations, and traffic routing rather than on end-device security mechanisms. With the absence of default NAT-based containment in IPv6, this assumption no longer holds. Depending on the network configuration, devices become directly reachable without any change to their internal security properties, thereby increasing their effective exposure [4], [5], [6]. The direct exposure of IoT devices is especially problematic due to the attack vectors commonly associated with this ecosystem. A large number of IoT devices have exposed services that are only protected by weak or default authentication mechanisms, enabling large-scale automated exploitation [7]. Embedded devices often contain insecure network services or misconfigurations, as well as persistent vulnerabilities produced by design constraints and limited update mechanisms [8]. Poorly secured management interfaces and API calls significantly expand the attack surface when IoT devices become externally reachable [9]. It is precisely the interaction between inherent device vulnerabilities and expanded reachability that increases the risk of remote exploitation.

1.3 Research Objectives The main objective of this thesis is to analyze how the transition to globally routable IPv6 addressing increases the attack surface of IoT devices by enabling direct external exposure in real-world network environments. Specifically, this study aims to identify under which router configurations, network parameters, and default device settings IoT devices become directly accessible from external networks.

1.4 Contributions This thesis contributes a reproducible residential IPv6 testbed for studying IoT exposure under controlled conditions, combining an ESP32 device running custom firmware, a residential ISP

router, an internal observer, and an external IPv6-enabled VPS. The design isolates the residential firewall state as the primary independent variable and operates on a real access network rather than an emulated environment. Using this setup, the thesis provides empirical evidence that, in the tested setting, external service exposure is determined primarily by the residential perimeter rather than by the device itself. It further distinguishes service accessibility from host observability, showing that a device may remain externally observable and hardware-identifiable even when the perimeter blocks inbound service traffic. The findings are then situated within the broader notion of security by obscurity, separating the architectural obscurity associated with IPv4/NAT from the computational obscurity associated with the IPv6 address space.

1.5 Thesis Structure Chapter2 provides the necessary technical background on IPv4, IPv6, residential network architectures, and IoT security models. Chapter3 describes the experimental methodology and testbed used to evaluate device exposure under different network configurations. Chapter4 presents the empirical results obtained across the evaluated scenarios. Chapter5 discusses these findings and examines their implications for IoT exposure in residential IPv6 environments. Chapter6 addresses the limitations of the study and discusses mitigation strategies at the network, device, and deployment levels. Finally, Chapter7 concludes the thesis by summarizing the main findings and outlining possible directions for future research.

2. Background and related work

2.1 The Loss of Perimeter Security The security of modern networks has been built around a well-defined perimeter, a topological barrier that separates trusted local networks (LAN) from wider, potentially hostile networks (WAN). However, the transition toward IPv6 alters this defensive boundary by removing legacy mechanisms and creating critical asymmetries in firewall policies. For decades, NAT protected IoT devices by defaulting to deny inbound traffic. However, with IPv6's direct global addressing, this implicit security disappears. As the research below demonstrates, organizations have failed to compensate with explicit firewall policies, creating a "perfect storm" of exposure [6], [10].

The elimination of NAT creates the need for IPv6 networks to implement filtering policies and firewalls that are equivalent to, or stronger than, those used in IPv4. Nonetheless, empirical research shows a substantial gap in the enforcement of these organizational policies. A large-scale characterization of dual-stack networks analyzing the behavior of 520,000 servers and 25,000 routers demonstrated that corporate security policies in IPv6 are consistently more permissive [10]. In particular, several high-value target applications were identified with more permissive security policies in IPv6. Management protocols commonly targeted by attackers and botnets, such as SSH, Telnet, and SNMP, were found to be more than twice as exposed on routers in IPv6 as they are in IPv4 [10]. This systemic weakness suggests that organizations are more inclined to prioritize securing legacy infrastructure while leaving IPv6 deployments insufficiently protected due to permissive configurations or technical and operational oversight [10].

While corporate networks already show significant security gaps, the situation in residential networks is more concerning. Recent studies on residential IPv6 deployments highlight significant security risks related with the residential transitions. Differently from IPv4 NAT, which by default imposes a default-deny access control model, IPv6 residential router manufacturers (CPE) have the discretion to decide whether to implement restrictive or permissive filtering policies. After actively scanning more than 2.5 million residential IPv6 prefixes, the researchers found that many routers adopted a default-allow policy [6], deliberately routing unsolicited incoming traffic from the WAN to the LAN. The study further showed that simple IPv6 scans allowed researchers to directly reach end devices such as printers, and smart light bulbs, surpassing the discovery rates that previously required exhaustive scanning of the entire IPv4 address space. These findings indicate that the residential transition weakens traditional firewall protections, exposing additional hosts to botnets and increasing overall Internet security risks [6].

2.2 The IoT Security Landscape The loss of the network perimeter described above conflicts with the current state of the embedded device ecosystem (IoT). The design of these devices often assumes that their operating environment will remain isolated and private, resulting in firmware architectures and communication protocols that are vulnerable when directly exposed to the Internet through IPv6.

Security practices in home IoT devices have been extensively evaluated [11]. After evaluating 45 commercial devices across four components: the device itself, the mobile application, the cloud endpoint, and the communication channel, this evaluation revealed that manufacturers often assume that the LAN environment is inherently trustworthy and protected by a boundary router. As a result, many devices rely primarily on the domestic router as the main security mechanism. This assumption appears as a consequence of prioritizing product release time over security considerations, which leads to the omission of internal security controls. Many devices lack local encryption mechanisms, making them susceptible to man-in-the-middle attacks, while mobile applications frequently request excessive privileges that go beyond their intended functionality [11]. These results are not limited to unbranded IoT devices. Platforms such as Samsung SmartThings also exhibit inherent design flaws in their permission model that result in over-privileged third-party applications, allowing potential remote exploitation when the devices are ex-

ternally accessible [9]. Considering these findings together with the disappearance of perimeter filtering policies, the implication is that devices and platforms designed under the assumption of network isolation become directly exposed to remote attackers.

In the application layer, the technical debt of the embedded ecosystem increases the viability of exploitation. Hardware limitations and accelerated development cycles lead to devices running obsolete or misconfigured network services. An examination of 32,000 firmware images discovered 38 previously unknown vulnerabilities in more than 693 firmwares, confirming that latent vulnerabilities persist due to outdated software and the fact that security is systematically sacrificed for cost and time-to-market [8]. The results are consistent with broader evaluations of domestic IoT devices [11], which revealed a persistent reliance on plaintext protocols such as HTTP instead of implementing TLS encryption, as well as the use of legacy administrative services such as Telnet. In traditional IPv4 networks behind NAT, interacting with these vulnerabilities would require prior access to the LAN. However, in the context of globally routable IPv6 networks, these insecure services can be directly exposed, facilitating credential extraction and the propagation of automated malware. The Mirai botnet demonstrated the consequences of this type of exposure, using a dictionary of only 62 pairs of default credentials for brute-force attacks via Telnet, the malware was able to infect more than 600,000 IoT devices and execute massive DDoS attacks against high-profile targets [7].

Even when manufacturers protect public web interfaces with passwords, devices may still expose residual risk through hidden administrative channels, as documented in routing analysis assessments [12]. Hidden network interfaces are defined as functional access routes to the device that are neither documented nor accessible through the standard navigation of the main portal, often introduced during development but later abandoned in production [12]. Because they are not exposed through the usual interaction flow, these interfaces evade standard authentication controls. A systematic analysis of 13 commercial devices discovered 79 hidden interfaces, 25 times more than previous tools commonly used. The activation of these hidden channels triggered 29 zero-day vulnerabilities, including command injection, XSS, information leakage, and operating backdoors that provided full access to the device, resulting in seven assigned CVEs [12]. The presence of this type of authentication-bypass vulnerability in embedded firmware has also been validated using binary analysis frameworks such as Firmallice, which detected backdoors that allowed administrative access through specially crafted packets without requiring legitimate credentials [13]. When these findings are considered together, the erosion of the local network perimeter not only exposes known vulnerabilities in IoT devices but also opens the path for potential system takeover and credential bypass.

2.3 IPv6 Visibility & Identifiability Even considering the scenario described above, a defensive argument that justified several of the shortcomings of the IoT ecosystem was that, regardless of its vulnerabilities, the 128-bit IPv6 address space provided a form of practical invisibility, in a role similar to NAT in IPv4, different in mechanism but often assumed to produce a comparable protective effect [14]. Nonetheless, core properties of IPv6, particularly its reliance on Stateless Address Autoconfiguration (SLAAC) without mandatory cryptographic authentication [5], [15] and the deterministic embedding of hardware MAC addresses via EUI-64 [16], make devices highly identifiable and enable the construction of precise visual and topological profiles for attackers.

In order to guarantee plug-and-play functionality, IPv6 [5] relies on Stateless Address Autoconfiguration (SLAAC) [17], a mechanism that uses mandatory messages in the Neighbor Discovery Protocol (NDP) [18], specifically Router Solicitations (RS) and Router Advertisements (RA). The structural weakness lies in the fact that the IPv6 standard does not require cryptographic authentication for these local network exchanges. The threats resulting from this absence were formally cataloged by the IETF [15], which documented how non-privileged actors within the same network segment can orchestrate man-in-the-middle attacks through falsified NDP messages, sending fake RA packets to reroute traffic or forcing denial of service during the autoconfiguration process. Even though Secure Neighbor Discovery (SEND) [19] was proposed as a mitigation mechanism, its implementation complexity and its architectural requirement for public keys have limited its practical deployment. The issue is further complicated because indiscriminately blocking

ICMPv6 [20] would break network connectivity, since NDP depends on these messages to operate [21]. As a result, the network initialization mechanisms used by IPv6, and therefore by many IoT devices, represent a permanently exposed attack surface.

The SLAAC process requires that the device generates its own interface identifier (IID) to use the last 64 bits of the IPv6 address. Traditionally, the industry has implemented the EUI-64 standard, which directly embeds the physical address (MAC) of the hardware into the public IID. The privacy and security implications of this design have been demonstrated through empirical measurements of EUI-64 address leakage [16]. These measurements revealed that this mechanism produces a systematic leakage of MAC addresses, allowing an unprivileged remote adversary to geolocate residential routers with street-level precision, identifying more than 12 million routers in 146 countries and territories with a median geolocation error of 39 meters [16]. By correlating the Organizationally Unique Identifier (OUI) with IPv6 geolocation databases and Wi-Fi positioning data, the investigation showed that an identifier at the network layer had become a direct vector for physical tracking and the identification of the device manufacturer. Once these identification capabilities are combined with firmware vulnerabilities (Section 2.2), a remote attacker could not only discover the IoT device but also determine the most likely manufacturer, firmware, and physical location, enabling highly targeted attacks.

The visibility of the connected ecosystem is further increased by the traffic generated at the application layer that modern protocols require to function properly. IoT devices use multicast protocols such as mDNS and SSDP for service discovery within the local network [22]. This passive traffic has been shown to form a highly descriptive network fingerprint [22]. Through the analysis of structural and textual information embedded in broadcast and multicast packets, the proposed framework was able to identify with high accuracy the brand, model, and category of IoT and mobile devices without actively interacting with them [22]. In an IPv6 environment where devices are directly reachable, this passive profiling capability complements manufacturer identification through EUI-64 [16], giving an attacker a detailed map of the device ecosystem before launching any exploit.

Summarizing, these findings destroy the illusion of opacity. In an IPv6 environment, an IoT device not only receives a publicly reachable address by default, but its autoconfiguration mechanisms also expose the identity of its manufacturer through EUI-64 addresses [16]. At the same time, NDP exchanges do not require mandatory cryptographic authentication [15], and broadcast protocols can passively reveal the device's brand, model, and category to any listener on the network segment [22]. As a result, the assumed "invisibility" of IoT devices disappears by design in the IPv6 environment.

2.4 Scanning & Target Generation in IPv6 The scanning and discovery phase is the first step in the chain of a cyberattack. Under IPv4, with an address space of only 2^{32} addresses, tools such as ZMap showed that it was possible to scan the entire Internet in less than one hour [23]. IPv6, with its much larger address space, was designed under the assumption that its virtually unlimited size would make exhaustive enumeration computationally impractical, turning address space size into a form of implicit security. However, recent studies have shown that this mathematical barrier has been systematically overcome through smart and adaptive scanning techniques.

The capability to scan the entire IPv4 space expanded large-scale Internet measurement. Building on tools such as ZMap [23], platforms like Censys allowed the indexing of vulnerabilities, certificates, and embedded devices across the Internet [24]. This visibility over the IPv4 address space made it possible to identify services and devices exposed at a global scale. Botnets such as Mirai exploited the same principle, locating IoT devices such as IP cameras by performing simple stochastic scans and compromising them through default Telnet credentials, as discussed in Section 2.2.

The IPv6 address space was designed under the idea that it would be too large to make exhaustive mapping computationally feasible. Nevertheless, recent research has shown that this space can still be mapped through adaptive scanning systems. One of the most notable contributions is *6Sense* [14]. One of the main challenges in IPv6 scanning is aliasing, where intermediate routers that are poorly configured respond to scans directed at tens of millions of inactive addresses within a prefix, diluting the resources of the scan-

ner with false positives [14]. 6Sense addresses this problem through a reinforcement learning approach combined with online and offline de-aliasing modules that can distinguish between illusory prefixes [14]. By dynamically learning active addresses and focusing on predictable assignment patterns—such as lower-order bytes used by IoT interfaces or grouping patterns in nybbles—the system was able to discover tens of millions of active IPv6 hosts, up to $3.6\times$ more active hosts than previous methods, and successfully correlate them with critical vulnerabilities (CVEs) [14]. Furthermore, side channels derived from ICMP rate limiting in routers allow the structure of IPv6 networks to be inferred without the need for direct scans [25]. These results establish that the virtually unlimited IPv6 address space is not a viable mechanism for obscurity.

2.5 Synthesis & Research Gap The analysis of the literature and related work presented in the previous sections reveals potential issues associated with the transition to IPv6. The previous sections examined four dimensions of this problem individually: the disappearance of the network perimeter within which the IoT environment was originally developed (Section 2.1), the inherent vulnerabilities of IoT devices (Section 2.2), the identifiability of devices in IPv6 networks (Section 2.3), and the feasibility of large-scale scanning in the IPv6 address space (Section 2.4). When considered in isolation, each of these phenomena represents a considerable risk. However, when examined together, a qualitatively different picture emerges. Despite this body of work, a gap in current research methodologies remains.

In a complete IPv6 environment, an IoT device connected to a residential network automatically receives a globally routable address, and in many cases the router does not filter incoming traffic [6], [21]. At the moment the device is autoconfigured through SLAAC, which relies on mechanisms that lack cryptographic authentication [15], the identity of the manufacturer may already be exposed through EUI-64 [16], while its discovery protocols may have already passively revealed its brand, model, and category [22]. At the same time, specialized scanning tools such as 6Sense can locate active devices within the 128-bit IPv6 address space [14]. Once discovered and identified, the device presents the same vulnerabilities extensively documented in the literature: firmware that runs unsafe protocols or lacks encryption [11], default credentials that can be exploited to obtain administrator privileges as demonstrated by the Mirai botnet [7], hidden interfaces that bypass authentication [12], and platforms with deficient permission hierarchies [9].

Each link of this chain has been individually validated, but the full chain remains unexplored. Network-focused studies concentrate on scanning feasibility and filtering policies without evaluating what happens once a device is reached. Device-focused research analyzes firmware vulnerabilities assuming an isolated IPv4 local network. No existing study examines, under controlled conditions, which specific router configurations and device settings enable or prevent the external reachability of an IoT device in a residential IPv6 environment.

This thesis addresses this gap through a controlled laboratory experiment using commercial hardware (ESP32), simulating being an IoT device, connected to real residential IPv6 infrastructure. By systematically varying router configurations and device parameters, this work identifies the specific conditions under which an IoT device with a globally routable IPv6 address becomes externally accessible, and demonstrates the security implications of that exposure.

3. Methodology

3.1 Research Design This chapter describes the methodology used to empirically evaluate the external reachability and service exposure of an IoT device in a residential IPv6 network configuration. The study follows a controlled case study design, in which a programmable embedded device (ESP32) is used as a representative IoT system. As discussed in Section 2.2, such devices commonly exhibit simplified network stacks and weak service configurations, making them suitable proxies for real-world IoT deployments. The device is systematically configured under different network conditions, while measuring its observability, accessibility, and network traffic visibility in each case. These configurations modify key parameters such as firewall state, access origin, and interaction protocols. The experimental setup is implemented in a physical laboratory environment using commercial hardware, including a residential router and an external VPS. This approach allows precise isolation of network variables while maintaining environmental control and reproducibility. At the same time, it avoids ethical concerns associated with scanning or interacting with third-party devices on the public Internet. The experimental design is structured around a set of progressive scenarios, ranging from an internal baseline to externally reachable IPv6 configurations and a final comparative scenario under IPv4. Each scenario is evaluated based on observable network and service interactions, including reachability at the ICMP level, TCP connectivity, and the presence of application-layer responses, allowing consistent expected results across repeated executions. All evidence is collected in raw form, including terminal outputs, packet captures, and device-level logs.

3.2 Testbed Architecture and Threat Model The testbed consists of four main components: an ESP32-based device, a residential router acting as the perimeter network controller, an internal host connected to the same LAN, and an external VPS located outside the residential network. This architecture allows the behavior of the embedded device to be observed from two complementary perspectives. The first was a local perspective, which made it possible to verify the internal availability of the IoT services and inspect traffic within the LAN, and, when communication between the embedded device and the internal host was not encrypted, to examine the content of the exchanged packets. The second was a remote perspective, representing potential access from outside the residential network.

Table 3.1: Testbed Components

Component	Role	Network Position	Notes
ESP32	Target IoT device	Residential WiFi; acquires global and link-local IPv6	Runs custom firmware; exposes HTTP (80) and Telnet (23)
Residential router	Perimeter controller	Provides native IPv6 and public static IPv4	Configurable IPv6 firewall (key independent variable)
Internal host	Local observer	Same LAN as the ESP32	Used for local validation and traffic inspection
External VPS	External host	Outside residential network; native public IPv6	Used for external interactions and IPv4 comparison

The target device was an ESP32 microcontroller executing custom firmware developed for this study.

Although the implementation is custom, its design follows the IoT service models and practices described in Section 2.2, rather than introducing new mechanisms. The firmware is implemented using the Arduino environment and the ESP32 network stack, including `WiFi.h`, `esp_event.h`, `esp_netif.h`, and the lwIP socket interface. These libraries enabled the management of the WiFi connection, network events, IPv6 initialization, and the creation of TCP services over IPv6 sockets. Once connected to the wireless network, the device automatically obtains both a link-local address and a global IPv6 address, and logs this information together with its MAC address through the serial output. This allows direct documentation of the device's network identity and addressing state within the experimental setup. At the application layer, the firmware implements two services over IPv6. The first is an HTTP server running on port 80, accessible through the main route `/`, which returns a simple dashboard with device information. Furthermore, the same server exposes an undocumented endpoint, `/hidden_admin`, reflecting a pattern commonly observed in IoT devices [12]. The second service is a Telnet-like TCP server on port 23, which requests credentials in plaintext and uses default credentials (`admin/admin`). Both services record inbound connections, including the source IPv6 address and remote port, as well as relevant events such as access to the hidden endpoint, authentication attempts, and executed commands. This enables the correlation between what is externally observable at the network level and the evidence generated locally by the device itself. The residential router is the perimeter component of the testbed. This device provides native IPv6 connectivity and a public static IPv4 address, which is essential for establishing a fair comparison with the IPv4 environment. In the case of IPv6, the router allows the firewall state to be modified explicitly, making this configuration one of the key independent variables of the experiment. In the case of IPv4, incoming connections are not directly possible by default unless an explicit port forwarding rule is configured. Since no such rule was enabled in this experiment, the IPv4 setup provides a useful baseline for contrasting both exposure models without introducing additional configuration elements that fall outside the scope of this study. Within this established perimeter, an internal host is also present. This host serves a double function. On the one hand, it is used to establish the local baseline of the experiment, meaning that it verifies that the device responds correctly at both the network and application layers before any external reachability test is performed. On the other hand, depending on the scenario, it acts as a passive observer, allowing traffic within the LAN to be captured and analyzed in order to verify whether the information exchanged via HTTP and Telnet remains visible in plaintext. The external VPS, located in a different country and equipped with native IPv6 connectivity, serves as the remote observation point and is used to generate ICMPv6 probes, TCP connection attempts, HTTP requests, and Telnet sessions from outside the residential network. From this architecture, two complementary observation modes are defined. The first corresponds to an active external observer, represented by the VPS, which attempts to reach the device through its global IPv6 address and evaluate its reachability, TCP connectivity, and response at the application layer. This same external position is also used in the final IPv4 comparison. The second corresponds to a passive local observer within the LAN. This distinction makes it possible not only to evaluate to what extent the device can be reached from outside the residential network, but also whether its communications remain visible inside the local network in the absence of encryption.

3.3 Experimental Scenarios The experiment is organized into five progressive scenarios (S0 to S4). Instead of evaluating each network parameter individually, the independent variables, such as the IPv6 firewall state, the origin of access, and the protocol used, are manipulated through these scenarios. In this way, each scenario represents a controlled condition under which changes in device reachability, service visibility, and traffic exposure can be observed.

Table 3.2: Experimental Scenarios

Scenario	Perimeter Condition	Access Origin	IP Context	Objective
S0 — Internal Baseline	not relevant	internal host	IPv6	confirm device identity, global IPv6 acquisition, and local availability of the implemented services
S1 — External Firewall ON	firewall enabled	external VPS	IPv6	evaluate whether the device remains observable despite TCP service filtering
S2 — External Firewall OFF	firewall disabled	external VPS	IPv6	evaluate whether the same global IPv6 address enables effective access to HTTP, Telnet, and the hidden endpoint
S3 — Passive LAN	not relevant	internal passive observer	IPv6	determine whether HTTP and Telnet communications remain visible in plaintext inside the LAN
S4 — IPv4 Baseline	no port forwarding	external VPS	IPv4	establish a contrast with residential IPv4 by evaluating whether equivalent direct inbound access exists

Scenario 0 establishes the internal baseline for the experiment. In this phase, the device is observed only within the LAN, with the objective of verifying that IPv6 connectivity is correctly initialized, that a global address has been acquired, and that the services implemented in the firmware respond as expected at both the network and application layers. This scenario allows us to ensure that any differences observed in subsequent phases are not due to device failures or misconfiguration, but rather the result of changes in the independent variables. Scenario 1 evaluates the behavior of the embedded device in the presence of an active external observer when the IPv6 firewall remains enabled. Under this configuration, the remote VPS attempts to reach the global address of the device through ICMPv6, TCP connections, and application-layer requests via the implemented HTTP and Telnet services. The objective of this scenario is to determine to what extent the device remains observable from outside the residential network, and what information can still be obtained despite the presence of firewall filtering. Scenario 2 maintains the same external origin and the same set of interactions, but with the IPv6 firewall disabled. This allows isolating the effect of changes in perimeter policies without modifying the firmware, the device, or the location of the observer. In this phase, it is evaluated whether the global address enables effective access to the HTTP and Telnet services, as well as to the hidden interface exposed through the endpoint `/hidden_admin`. Scenario 3 changes the observation perspective to a local context and uses an internal host as a passive observer. Differently from the previous scenarios, the goal here is not to measure remote accessibility, but rather traffic visibility. This scenario allows verifying whether the information exchanged with the device, including

HTTP requests, server responses, Telnet credentials, and executed commands, remains visible in plaintext when no encrypted communication is used, as commonly observed in legacy IoT services such as Telnet [7]. The final scenario (S4) introduces a reference condition under IPv4. In this phase, the same external VPS attempts to access the device through a public and static IPv4 address, while maintaining the absence of port forwarding rules. The objective of this scenario is to evaluate whether the exposed services can be reached under typical residential IPv4 conditions, or whether they remain effectively isolated due to Network Address Translation (NAT), which restricts unsolicited inbound connections unless explicit port forwarding is configured.

3.4 Measurement and Evaluation The evaluation of each scenario will be done through three complementary levels: reachability at the network layer, the accessibility of the services, and the traffic visibility at the application layer. This distinction is important because it reflects different stages of exposure within the experiment. Reachability at the network layer is evaluated through ICMPv6 probes and, in the last scenario, also through IPv4 tests, in order to verify whether the device responds under the tested perimeter configuration. Service accessibility is evaluated through TCP connection attempts and application layer interactions, including port scanning, HTTP requests, and Telnet sessions. Finally, traffic visibility is evaluated through packet captures and logs generated by the device itself, using tools such as Wireshark to inspect the exchanged traffic and determine whether information such as HTTP requests, server responses, credentials, and commands remains observable in plaintext. For this purpose, standard network analysis tools are used. Reachability tests are performed using `ping6` and `ping`, the state of TCP services is observed with `nmap`, interaction with the HTTP server is validated using `curl`, and Telnet sessions are documented through connection transcripts. At the traffic level, `tcpdump` is used, and when needed, captures are analyzed in Wireshark to inspect packet contents. These external observations are complemented with the ESP32 serial output and the logging mechanisms implemented in the firmware, which record inbound connections, requested resources, authentication attempts, and executed commands. Within this work, a device is considered reachable when it produces valid responses at the network level under the tested configuration. A service is considered accessible when a TCP connection can be established and a valid response is obtained at the application level. In the same way, traffic is considered visible when the exchanged content can be directly inspected in the captures without the need for decryption. These criteria are applied consistently across all scenarios, so that the observed changes can be attributed to the experimental conditions and not to differences in the measurement method.

Table 3.3: Measurement Dimensions and Evidence Sources

Measurement Dimension	Operational Evidence	Primary Tools / Sources
Network reachability	valid ICMP responses under the tested configuration	ping6, ping, serial logs
TCP service state	port state observed as open, filtered, or closed/unreachable	nmap, TCP connection attempts
HTTP accessibility	valid HTTP response received from the root endpoint or hidden endpoint	curl, HTTP output, serial logs
Telnet accessibility	successful session establishment and application-layer interaction	Telnet transcript, serial logs
Traffic visibility	readable application-layer content visible in packet captures without decryption	tcpdump, Wireshark
Device-side logging evidence	locally recorded evidence of inbound connections, endpoint access, and authentication attempts	ESP32 serial output, firmware logs

3.5 Ethical Considerations The experiments are conducted within a controlled infrastructure. The experimental setup is limited to a residential network, an external VPS under the researcher’s control, and an ESP32 device with firmware developed specifically for the purposes of this work. No third-party devices, production systems, or publicly accessible IoT deployments are included in the testbed. The design does not include internet-wide scanning or any interaction with systems outside the experimental environment. Traffic inspection is limited to communications involving the test device and researcher-controlled hosts, keeping the methodology within a closed and ethically controlled setup.

4. Experimental Results

4.1 Address Assignment and Identifier Exposure Upon connecting to a residential WiFi network, the ESP32 successfully completed the IPv6 autoconfiguration process through SLAAC. The logs confirm the assignment of a link-local address as well as a global IPv6 address, together with the MAC address and the parameters observed in Table 4.1.

Table 4.1: Device Identity Parameters

Parameter	Value
MAC address	28:05:A5:0E:73:6C
Link-local IPv6	FE80::2A05:A5FF:FE0E:736C
Global IPv6	2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
Address acquisition method	SLAAC (Stateless Address Autoconfiguration)
IID generation method	EUI-64

The link-local address was assigned first, followed by the global address after the reception of the Router Advertisement. Both addresses coexisted in the interface during all the experimental scenarios. The interface identifier (IID) of the global address follows the EUI-64 format, embedding the device's MAC address (28:05:A5:0E:73:6C) directly into the IPv6 address [16] (Table 4.1).

4.2 Internal Baseline and Local Service Availability (S0) The ESP32 responds correctly to ICMPv6 probes, and both TCP ports (80 and 23) are reachable, with the implemented services behaving as expected (Table 4.2).

Table 4.2: S0 Internal Baseline Results

Test	Tool	Result
ICMPv6 reachability	ping6	4/4 replies; initial RTT spike, then stabilized around 7 ms
Port 80 state	nmap -6 -sT	open
Port 23 state	nmap -6 -sT	open
HTTP root (/)	curl	HTTP 200 OK (MAC + IPv6 information)
HTTP hidden endpoint (/hidden_admin)	curl	HTTP 200 OK (Diagnostic Mode: ENABLED)
Telnet session	ncat	credentials transmitted and accepted in plaintext

Accessing the HTTP root endpoint returns a valid response with device information, and the undocumented endpoint /hidden_admin is also accessible without authentication (Table 4.2). Furthermore, a Telnet session can be established using default plaintext credentials. All interactions were recorded in the ESP32 serial logs, which capture source addresses, requested resources, and authentication events (see Appendix B.1 for the raw logs of this scenario). This confirms that both the device and its services were fully

operational before introducing any external interaction.

4.3 External IPv6 Reachability with Firewall Enabled (S1) ICMPv6 probes received responses in all attempts (10/10), with an average RTT of 40 ms (Table 4.3).

Table 4.3: S1 Results: External IPv6, Firewall Enabled

Test	Tool	Result
ICMPv6 reachability	ping6 -c 10	10/10 replies, 0% loss, avg RTT 40 ms
Port 80 state	nmap -6 -sT	filtered (no-response)
Port 23 state	nmap -6 -sT	filtered (no-response)
HTTP root (/)	curl	connection timed out after 10 s
HTTP hidden endpoint (/hidden_admin)	curl	connection timed out after 10 s
Telnet session	ncat	no connection established

However, no transport-layer connectivity was observed. Ports 80 and 23 were reported as filtered, with no TCP responses (Table 4.3). HTTP requests timed out, and no Telnet session could be established. The device did not register any incoming TCP connections during this scenario. Packet captures on the VPS confirmed that only ICMPv6 traffic reached the destination, while all TCP SYNs remained unanswered (supported by logs in Appendix B.2).

4.4 External IPv6 Accessibility with Firewall Disabled (S2) ICMPv6 connectivity remained consistent, with 10/10 responses and an average RTT of 40 ms (Table 4.4).

Table 4.4: S2 Results: External IPv6, Firewall Disabled

Test	Tool	Result
ICMPv6 reachability	ping6 -c 10	10/10 replies, 0% loss, avg RTT 40 ms
Port 80 state	nmap -sT	open (syn-ack)
Port 23 state	nmap -sT	open (syn-ack)
HTTP root (/)	curl	HTTP 200 OK (MAC + IPv6 exposed)
HTTP hidden endpoint (/hidden_admin)	curl	HTTP 200 OK (Diagnostic Mode: ENABLED)
Telnet session	ncat	credentials transmitted and accepted in plaintext
Plaintext exposure	tcpdump -A	full HTTP request and response observable in plaintext

Ports 80 and 23 were reported as open, and both services were accessible from the external host (Table 4.4). The HTTP root endpoint returned a valid response, the /hidden_admin endpoint was accessible without authentication, and the Telnet session accepted the default plaintext credentials. HTTP responses included device information, and the logs confirmed incoming connections from the external address along with the associated events (see Appendix B.3). Traffic captures revealed that the HTTP exchange, including the request to /hidden_admin and its response, was completely visible in plaintext.

4.5 Local Traffic Visibility and Plaintext Exposure (S3) Traffic captures allowed direct observation of HTTP requests, full server responses, Telnet authentication prompts, and credentials in plaintext (Table 4.5).

Table 4.5: S3 Results: Local Passive Observation

Observation	Tool	Result
HTTP request	Wireshark	full HTTP request observable in plaintext
HTTP response	Wireshark	full HTTP response observable in plaintext
Telnet banner	Wireshark	Password prompt visible in ASCII payload
Telnet credential	Wireshark	credentials transmitted and accepted in plaintext
MAC address	Wireshark	MAC 28:05:A5:0E:73:6C in frame header

The device's MAC address was visible in the Ethernet headers (Table 4.5). Both HTTP and Telnet traffic could be inspected directly without the need for decryption.

4.6 IPv4 Baseline Without Port Forwarding (S4) The router responded to ICMP probes, confirming network-level connectivity (Table 4.6).

Table 4.6: S4 Results: IPv4 Baseline

Test	Tool	Result
IPv4 reachability	ping -c 5	5/5 replies from router, avg RTT 35 ms
Port 80 state	nmap -sT	filtered (no-response)
Port 23 state	nmap -sT	filtered (no-response)
HTTP root (/)	curl	connection timed out after 10 s

Ports 80 and 23 were reported as filtered, and HTTP requests timed out with no response (Table 4.6). No direct inbound access to the device's services was observed under this configuration.

4.7 Cross-Scenario Summary The results show consistent differences between the internal baseline (S0), external network-level response without application access (S1), external service accessibility (S2), local traffic visibility (S3), and the IPv4 scenario without port forwarding (S4), as summarized in Table 4.7.

Table 4.7: Cross-Scenario Comparative Results

Metric	S0 Int.	S1 Ext.	S2 Ext.	S3 Local	S4 IPv4
Global IPv6 assigned	✓	✓	✓	✓	—
EUI-64 MAC derivable	✓	✓	✓	✓	—
ICMP response rate	100%	100%	100%	—	—
Mean RTT	7 ms	40.8 ms	40.4 ms	—	35.6 ms
Port 80 state	open	filtered	open	—	filtered
Port 23 state	open	filtered	open	—	filtered
HTTP root accessible	✓	✗	✓	—	✗
Hidden endpoint accessible	✓	✗	✓	—	✗
Telnet session established	✓	✗	✓	✓	✗
Plaintext payload visible	—	—	✓	✓	—
Credentials visible	—	—	✓	✓	—
Inbound TCP connectivity	✓	✗	✓	✓	✗

The results presented in this section document the behavior of the device under the different perimetral conditions (Table 4.7). The observations focus on connectivity, service exposure, and identifier visibility, and serve as empirical evidence for the discussion that follows.

5. Discussion

The results presented in the previous chapter allow the identification of consistent patterns in the interaction with IoT devices under different network configurations. In this chapter, these results are interpreted in relation to the research objectives and the background presented earlier, highlighting the role of the network perimeter, the reachability of the device under IPv6, and the conditions under which its services become externally accessible.

5.1 Perimeter-Dependent Exposure as the Primary Risk Condition A consistent observation across the experiment is that the external reachability of the device is determined not by the device itself, but by the filtering policy applied by the residential firewall. Between S1 and S2, the ESP32 maintained the same global IPv6 address, the same service configuration, the same default credentials, and the same firmware version. The only parameter that changed was the IPv6 residential router firewall state. Nonetheless, this change in the perimeter configuration resulted in two completely different exposure states.

As summarized in Table 4.7, the cross-scenario comparison makes this asymmetry explicit. The ICMP response rate remains at 100% and the EUI-64-derived MAC address is consistently observable across both scenarios, while port state, service accessibility, and inbound TCP connectivity differ entirely between S1 and S2. With the firewall enabled, ports 80 and 23 appeared as filtered, HTTP requests expired, and it was not possible to establish any Telnet session. Furthermore, the serial logs did not show any incoming TCP connections, indicating that the traffic was discarded before reaching the application layer. Once the firewall was disabled, both ports were reported as open, the dashboard and the hidden admin endpoint became reachable, and the Telnet service accepted the default credentials "admin/admin". As there was no modification in the host or its services, the observed differences can be attributed to the independent variable of the perimeter policy applied by the router.

The contrast observed between these scenarios indicates that, in residential IPv6 environments, exposure depends primarily on the perimeter configuration rather than representing an inherent vulnerability of IPv6 itself. Although IPv6 restores the architectural possibility of end to end connectivity through globally routable addressing, whether a device becomes externally reachable still depends on the filtering behavior of the residential router. The architecture enables the communication path, but the perimeter policy determines whether that path remains accessible from the Internet.

With this in mind, the expected security role of a residential IPv6 router is to enforce a restrictive default policy for unsolicited inbound traffic, particularly for TCP connections [6]. The difference between S1 and S2 illustrates the practical meaning of this requirement. Under the restrictive policy, the device remained globally addressed but service inaccessible from the outside. When that filtering function was removed, the same address became a functional entry point to the device's exposed services.

This result is also relevant at larger scales beyond the residential setting. Prior measurements of dual stack networks have shown that IPv6 filtering policies can be more permissive than their IPv4 counterparts, including for high value services such as SSH, Telnet, and SNMP [10]. This suggests that the configuration gap observed in the experiment reflects a broader operational problem, where IPv6 connectivity is deployed without an equivalent security policy. Within the residential testbed, this condition was sufficient to expose weak internal services without requiring any modification to the device itself.

The key point is therefore not that IPv6 creates new vulnerabilities in the device itself, but that global addressing introduces an exposure vector through which pre-existing vulnerabilities become externally exploitable.

5.2 ICMPv6 Reachability as a Persistent Discovery Vector The previous finding establishes the residential firewall as the decisive factor in determining whether global addressability translates into exposure at the application layer. However, S1 also reveals a more subtle distinction, blocking access at the application layer is not equivalent to eliminating external observability. Even when the firewall successfully prevented all inbound TCP connectivity, the device remained externally observable. In S1, HTTP requests expired, no Telnet session could be established, and the device registered no incoming application-layer connections. Nonetheless, the same device responded consistently to ICMPv6 probes sent from the external VPS, with 10/10 valid replies, zero packet loss, and stable RTT values (Table 4.3). In S2, where TCP access was permitted, ICMPv6 behavior remained practically identical (Table 4.4). The contrast between both scenarios shows that the firewall policy modified service accessibility, but not the ability of an external observer to verify that the host was active and reachable at the network layer.

This behavior is not necessarily the result of a misconfigured router, but rather a consequence of how IPv6 networking operates. ICMPv6 is not an auxiliary protocol that can be indiscriminately blocked without affecting network functionality. Unlike the common treatment of ICMP in IPv4 environments, IPv6 depends on ICMPv6 for normal operation. Neighbor Discovery, router discovery, and SLAAC rely directly on ICMPv6 exchanges, and the same process through which the ESP32 acquired its global IPv6 address depends on these mechanisms. Blocking ICMPv6 indiscriminately at the network boundary would interfere with normal IPv6 operation, including connectivity diagnostics and path MTU discovery, which is why selective filtering that preserves essential protocol functionality is the recommended approach [21]. From this perspective, the fact that the device remained pingable even behind a correctly configured firewall does not necessarily indicate misconfiguration, but rather a predictable consequence of a functional IPv6 deployment.

This result becomes especially relevant when considering the addressing properties discussed in Section 4.1. The global IPv6 address assigned to the ESP32 embeds the MAC address directly in the interface identifier through EUI-64, making the address stable and manufacturer-identifiable [16]. As a result, an external observer receiving a valid ICMPv6 reply not only confirms that the host is active, but can also derive the hardware address and identify the manufacturer through the OUI prefix without requiring additional interaction. Under S1 conditions, the exposed services remained inaccessible, but the host itself was still externally identifiable through a stable, hardware-derived network identity.

S1 reveals a separation between two conditions that are often treated as equivalent, discoverability and exploitability. The firewall reduced the attack surface by preventing access to HTTP and Telnet, but it did not remove the device from external visibility. If an external actor already knows the IPv6 address of the host, or is able to infer it through target generation techniques, a single ICMPv6 probe is sufficient to confirm that the system exists, responds, and remains active. In this sense, the firewall operates as a barrier against direct exploitation, but not as a mechanism that eliminates host discoverability. Candidate IPv6 addresses generated through adaptive target-generation techniques can still be validated through probing even when service-level access remains blocked [14].

S1 therefore represents an observable host whose services remain inaccessible from outside the residential network. S2, by contrast, represents the same host under a perimeter configuration where access to the application layer becomes possible. The difference between both scenarios is not the presence of the device in the network, but whether that presence translates into effective interaction with its exposed services. The residential firewall successfully blocked direct access to HTTP and Telnet, but it did not eliminate the external visibility of the host. It should therefore be understood as a necessary mechanism for containing direct exploitation, but insufficient for eliminating external detectability. This gap between what the firewall prevents and what it is often assumed to prevent connects directly with the broader conditions examined in the following section.

5.3 The End of Security by Obscurity Residential IoT deployments have often relied on the assumption of limited direct reachability from the public Internet. This assumption results from two independent processes under different protocol conditions, each associated with a different form of obscurity. In IPv4,

obscurity is architectural and produced by NAT, where devices do not have publicly routable addresses. In IPv6, obscurity is assumed to be computational, based on the size and structure of the address space, where devices are globally addressable but expected to be difficult to discover.

Both forms of obscurity are reflected in scenarios S2 and S4. Under IPv4 (S4), the external VPS reached the public address of the residential router, but no service on the ESP32 was accessible. All device functionality remained confined to the local network. Under IPv6 (S2), the same device became directly reachable once the firewall was disabled, exposing HTTP endpoints and a Telnet service without any modification to the device itself, as shown in Table 4.4 and Table 4.6. The difference is not the device, but the conditions under which it is exposed.

Under IPv4/NAT, this containment is produced by the network architecture itself. Since no mapping exists for unsolicited inbound traffic, inbound connections are discarded by default [2]. Under IPv6, this implicit containment disappears once the device acquires a globally routable address. From that point, whether the services remain reachable from the Internet depends on the filtering policy applied by the residential router. As observed in S2, removing that filtering policy was sufficient to expose all running services without any modification to the device itself.

A second form of obscurity is associated with the size and structure of the IPv6 address space. Even though IPv6 devices are globally addressable, it is often assumed that large scale host discovery remains impractical due to the size of the address space itself. The observations discussed in Section 5.2 show that this assumption does not fully hold. Even when service-level access remained blocked, the device continued responding to ICMPv6 probes, while the EUI-64 addressing mechanism introduced predictable structure into the interface identifier [16]. At the same time, adaptive target-generation techniques have demonstrated that active IPv6 hosts can still be discovered through learned address patterns, while network structure can be inferred through side-channel methods without exhaustive enumeration [14], [25]. In this sense, the size of the IPv6 address space alone is not sufficient to prevent external discovery.

The results discussed throughout this section show that neither form of obscurity remains sufficient under the conditions examined in this thesis. The device is no longer indirectly protected by the absence of a publicly reachable address, nor can it rely on address-space size alone to remain difficult to discover. The transition examined here is therefore not a change in the internal security properties of the device itself, but the progressive removal of external conditions that had historically limited its exposure. This is the shift reflected in the title of the thesis, from security by obscurity to imminent exposure.

5.4 Weak Services as Exposure Amplifiers Once the device becomes externally reachable, the behavior of the exposed services becomes directly relevant to the security state of the system. The experiment does not only evaluate whether the ESP32 can be reached from outside the residential network, but also what the device exposes once that access becomes possible. Under these conditions, the implemented HTTP and Telnet services offered no meaningful resistance once inbound traffic reached the device.

These weaknesses were deliberately implemented for the purposes of the experiment, but they were not arbitrary. S2 confirmed that all three became directly accessible once inbound traffic was permitted (Table 4.4, Appendix B.3). As discussed in Chapter 2, IoT security research has repeatedly documented devices that expose weak application-layer controls, provide management interfaces over HTTP, retain legacy administrative services such as Telnet, or include undocumented functionality outside the normal user interface. The firmware used in this experiment does not attempt to reproduce every possible IoT device, nor to emulate a specific commercial product. Instead, it implements controlled weakness patterns that have already been observed in real embedded deployments [8], [11], [12]. The corresponding firmware implementation, including the HTTP and Telnet service logic, is documented in Appendix A.

The Telnet service represents the clearest example of this weakness pattern. Plaintext administrative access and default credentials have repeatedly appeared in large scale IoT compromise, most notably in the Mirai botnet [7]. The credential pair `admin/admin` used in this experiment reflects the same type of weak authentication mechanism exploited in those attacks. S3 additionally showed that this weakness was not limited to remote access. Within the local network, the Telnet authentication exchange, including

prompts and credentials transmitted in plaintext, was directly observable in Wireshark captures without any decryption process (Table 4.5 and Appendix B.4).

The broader implication is that exposure does not create the weakness itself, but changes the conditions under which that weakness becomes relevant. Under the IPv4 conditions observed in S4, plaintext Telnet, weak authentication, and the hidden administrative endpoint remained confined to the local network. Under S2, the same services became directly reachable from outside the residential environment. The device itself did not become less secure between scenarios. What changed was whether those weaknesses remained locally contained or became externally accessible through the network perimeter.

For this reason, weak services act as exposure amplifiers. The permissive IPv6 perimeter does not create the insecure HTTP endpoint, the hidden administrative interface, or the default Telnet credentials. What changes is the scope under which those weaknesses operate. Services that would otherwise remain locally contained become remotely reachable once the device acquires a globally routable address and the residential router permits inbound traffic.

5.5 Validity and Scope of the Findings The experimental results are supported by a set of conditions that remained consistent across all scenarios. Between S1 and S2, the firewall state was the only parameter that changed, while the device, firmware version, service configuration, and measurement tools remained unchanged throughout the experiment. S0 fulfilled its role as an internal baseline by verifying that the HTTP and Telnet services were operational before any external interaction was introduced. This condition is important because the contrast between S1 and S2 depends on the device remaining functional before the perimeter configuration changed. The observed differences between both scenarios can therefore be attributed to the residential filtering policy rather than to inconsistencies at the device level. Additional validation was provided through the ESP32 serial logs, which recorded inbound connections, requested resources, and authentication events across the different scenarios. This allowed the external observations to be compared directly with the events registered by the device itself. The measurement criteria defined in Section 3.4, including ICMPv6 reachability, TCP port state, HTTP and Telnet accessibility, and traffic visibility, were applied consistently throughout the experiment, and the observations remained coherent across the collected evidence within each scenario.

Several constraints limit the external validity of the experiment. First, the evaluation was conducted using a single embedded device running custom firmware on the lwIP network stack. Although the implemented services and configuration patterns were intentionally based on weakness patterns documented in the IoT literature (Chapter 2), the observations cannot be assumed to represent all firmware implementations, device categories, or hardware architectures. Second, the residential router used in the experiment corresponds to a specific model provided by a Swiss ISP. The observed firewall behavior, including the fact that ICMPv6 probes received valid replies while inbound TCP connections remained blocked in S1, reflects the implementation decisions of that particular device. Different residential routers may apply different default filtering policies, stateful inspection behavior, or ICMPv6 handling under equivalent configurations. Third, all external measurements were performed from a single VPS located in a European datacenter and during a limited observation period. The measured RTT values and connectivity behavior therefore reflect that specific routing context and may differ under alternative network paths, geographic locations, or upstream filtering conditions.

Although the experimental setup reflects a realistic and reproducible residential IPv6 deployment using real commercial hardware and infrastructure, the results should be interpreted as evidence of a broader class of risk conditions rather than as a universal characterization of all residential IPv6 environments. More specifically, the experiment examines how perimeter configuration influences the external exposure of IoT services under globally routable addressing. Within this scope, the central observation is not that all IPv6 residential networks expose IoT devices, but that under the tested conditions a single change in the residential filtering policy was sufficient to move the same device from local availability to external service accessibility.

6. Limitations and Mitigation

6.1 Limitations Beyond the methodological limitations discussed in Section 5.5, several broader limitations remain. These do not affect the internal consistency of the observations reported in Chapter 4, but rather delimit the scope under which the findings should be interpreted.

The residential firewall was evaluated under two states only, enabled (Scenario S1, Section 4.3) and disabled (Scenario S2, Section 4.4). This was enough to isolate the perimeter as the primary exposure variable, but it does not capture the intermediate filtering policies that may be present in residential environments, such as selective TCP filtering, ISP-defined configurations, per-source rate limiting, or stateful inspection rules tied to outbound flows [6]. The contrast summarized in Table 4.7 therefore reflects the behavior of a real residential deployment under two specific perimeter conditions, but should not be interpreted as covering the full range of filtering behavior that may exist across residential environments.

The external VPS reached the ESP32 by using the global IPv6 address recorded directly from the device serial output (Section 4.1, Appendix B.1), meaning that the host address was already known before the external interaction began. The experiment therefore evaluates reachability under known-address conditions rather than the process through which a host becomes discoverable. Whether the same device would have been identified through adaptive scanning techniques [14] or inferred through side-channel observations [25] under the same residential configuration was not evaluated within the experimental setup, although both approaches were discussed in the literature reviewed in Section 2.4. The observations therefore describe the conditions under which external access becomes possible after host identification, rather than the complete path from discovery to service access.

The scope of the threat model (defined in Section 3.2) remains bounded to three observable conditions, network reachability, service accessibility, and traffic visibility. The scenarios showed that the HTTP root endpoint, the hidden administrative interface, and the Telnet service became directly reachable from the external VPS under Scenario S2 (Section 4.4), and that authentication with the default credentials succeeded (Table 4.4, Appendix B.3). However, the experiment did not extend beyond the point of successful access toward actions such as persistence, lateral movement, or remote code execution. The observations therefore describe the conditions that enable an external host to reach the application layer, rather than the complete impact that may follow after access has been obtained. Prior work has documented the consequences of comparable access conditions at larger scale [7], [11], [12], whereas the present experiment evaluates the entry condition rather than the full sequence of exploitation events.

The experiment also remains bounded to a specific IPv6 deployment model. The IPv4 comparison performed in Scenario S4 (Section 4.6) uses a native dual-stack residential connection, in which the same router provides a public static IPv4 address alongside native IPv6 (Section 3.2, Table 3.1). The experiment does not address alternative deployment models such as carrier-grade NAT combined with IPv6, IPv6-only access with NAT64, or tunneling-based transition mechanisms. These configurations may introduce different perimeter conditions and should not be assumed to reproduce the IPv6/IPv4 contrast observed in this work.

6.2 Network-Level Mitigations Within the scope defined above, the clearest mitigation target is the residential perimeter. Although the experiment does not characterize all possible IPv6 deployment models or filtering policies, the S1/S2 contrast (Section 4.3 vs. Section 4.4, see Table 4.7 and Section 5.1) shows that the firewall state alone was enough to move the same device from external service inaccessibility to full external reachability. Mitigation at this layer is therefore the most immediate way to reduce exposure, since it does not require any modification of the device, the firmware, or the IPv6 protocol itself.

The first and most immediate mitigation concerns the filtering policy applied at that perimeter. The

expected baseline behavior of a residential IPv6 router is to discard unsolicited inbound traffic by default and to allow it only in response to explicit configuration or to outbound-initiated flows [26]. This behavior is shown in Scenario S1 (Section 4.3), where the router blocked inbound TCP traffic toward ports 23 and 80 despite the device being globally addressed (Table 4.3, Appendix B.2). However, the observations suggest that the relevant objective is not simply enabling or disabling the firewall as a whole, but controlling which devices and services remain externally reachable. A device intended to be accessible from outside the residential network should be exposed through a specific and deliberate rule, in the same way that an IPv4 deployment exposes an internal host through an explicit port-forwarding entry rather than by removing NAT entirely (Section 3.2). Under IPv6, this selective exposure remains architecturally possible and forms part of the recommended residential security model [26], but it depends on the availability of filtering controls at the level of individual devices and services rather than a single global switch.

The practical difficulty is that this level of granularity is not always available. Measurements of residential deployments document variations in default configurations and inbound filtering behavior [6]. Similar differences have also been observed in organizational networks, where IPv6 policies remain more permissive than their IPv4 counterparts even for protocols such as SSH and Telnet [10]. This observation is particularly relevant because filtering behavior in IPv4 should not be assumed to transfer automatically to IPv6. This is visible in the experiment itself: the same services remained unreachable under IPv4/NAT (Scenario S4, see Table 4.6) but became directly accessible under IPv6 once the corresponding filtering policy was not enforced (Scenario S2, see Table 4.4). When the only available control consists of a binary firewall state, selective exposure becomes difficult to achieve. The perimeter either remains fully restrictive, which also blocks legitimate inbound services, or becomes broadly permissive, reproducing the conditions observed in Scenario S2. In this context, the responsibility extends beyond the operator of the device itself and depends heavily on the default configurations and control mechanisms provided by router manufacturers and ISPs.

A second consideration applies to the treatment of ICMPv6 at the network boundary. The observation discussed in Section 5.2 that the ESP32 remained pingable under Scenario S1 does not imply that all ICMPv6 traffic should be discarded. ICMPv6 forms part of the normal operation of IPv6 itself, since Neighbor Discovery, address resolution, and Path MTU Discovery depend on it [18], [20], and indiscriminate filtering would interfere with the same SLAAC process through which the ESP32 acquired its global address (Section 4.1). A more appropriate approach is selective filtering, where message types required for protocol functionality are preserved while non-essential ICMPv6 traffic originating from outside the network can be filtered or rate-limited [21]. Under such conditions, the simplest mechanism for externally confirming host activity can be reduced, although adaptive scanning and indirect inference techniques may still identify active hosts without relying on echo responses [14], [25].

6.3 Device-Level Mitigations While the previous paragraph addresses how external traffic reaches the device, the present section focuses on what the device reveals about itself once it participates in IPv6 communication. The two operate on different dimensions of exposure. The first determines whether the device can be reached, while the second determines what remains observable even when service access is restricted. The findings presented in Section 4.1 show that the ESP32 derived its interface identifier directly from its hardware MAC address through the EUI-64 mechanism, producing a stable and manufacturer-identifiable global IPv6 address (Table 4.1). This identifier remained unchanged across all scenarios and remained observable even under the most restrictive perimeter condition tested, since the ICMPv6 responses recorded in Scenario S1 (see Appendix B.2) carried the same EUI-64-derived address despite all TCP traffic being blocked. As discussed in Section 5.2, this means that perimeter filtering alone does not prevent the disclosure of hardware-derived addressing information. An external observer holding a candidate address may still infer manufacturer-attributable information regardless of whether service-level access is available [16]. Application-layer weaknesses such as those examined in Section 5.4 require their own firmware-level mitigations, but these concerns are not specific to the IPv6 transition addressed by this work and have already been extensively documented elsewhere [8], [11].

The mitigation therefore lies at the addressing layer of the device itself. Two complementary mecha-

nisms have been standardized. Stable but semantically opaque interface identifiers preserve addressability without embedding hardware-derived information [27], while temporary interface identifiers rotate over time and add temporal unlinkability to outbound communications [28]. The two are not mutually exclusive, since a device may use a stable opaque address for incoming connections while generating temporary addresses for outbound flows. The lwIP-based stack used in this work implements neither mechanism and falls back to EUI-64 by default, which produced the address structure observed in Table 4.1. The mitigation therefore depends on the firmware author and on the network stack they choose. The residential perimeter cannot modify how interface identifiers are generated, and application-layer hardening does not alter addressing decisions made beneath it.

6.4 The Limits of End-User Responsibility The mitigations discussed previously in this chapter are addressed primarily to ISPs, router manufacturers, and IoT device manufacturers. This allocation reflects the fact that the variables that determined the exposure observed in this work were largely outside the practical control of the end-user of the device.

This limitation begins with what the user can observe. From inside the residential network, the device may appear to behave identically whether the perimeter is filtering external traffic or not. The firewall state changes what an external observer sees (Scenario S1 vs. Scenario S2, see Section 5.1, Table 4.7), but it does not necessarily change how the device behaves toward local hosts. An end user without access to external infrastructure therefore has no straightforward way to determine whether their device is reachable from the public Internet or what it exposes there. Measurements of the residential IPv6 ecosystem confirm that permissive default policies occur in real CPE configurations [6], but the user has no built-in mechanism to verify which configuration applies to them.

Even when the situation is recognized, the controls available to the user do not match the structure of the exposure. External service reachability, ICMPv6-level discoverability, and EUI-64-based identifiability arise from different mechanisms and remain present under different conditions (Section 5.1 and Section 5.2). The binary IPv6 firewall control exposed by the testbed router reduces these dimensions to a single decision and does not address EUI-64 identifiability at all, since the addressing behavior of the device is determined by the network stack and remains outside any policy applied at the perimeter.

What remains within the user's reach is bounded and indirect. Keeping firmware updated, avoiding the explicit activation of features that expose the device beyond the local network when not required, and isolating IoT devices on a separate segment when the router supports it are reasonable practices, but they would not have altered the S1/S2 contrast (Section 4.3 vs. Section 4.4). The decisive variables remained in the firmware and in the router defaults, and ordinary user action could not directly change either layer.

This does not mean that end users are exempt from operational responsibility, but that their position is structural. They inherit an exposure baseline they did not design, they lack the visibility to evaluate it, and they lack the controls to refine it. Any framework that intends to reduce the exposure observed in this work across the broader residential ecosystem must therefore begin with the actors that determine those baselines, in line with the mitigations discussed for network operators (Section 6.2) and device manufacturers (Section 6.3).

7. Conclusion

This thesis examines how the transition to globally routable IPv6 addressing affects the exposure of IoT devices in residential networks, and specifically under which router configurations and device settings such a device becomes directly accessible from outside the residential network. To answer this question, a controlled case study was conducted in which a single embedded device, with fixed firmware, service configuration, and addressing behavior, was observed under different perimeter and protocol conditions. This design allowed changes in external reachability to be attributed to the experimental variable rather than to changes in the device itself. The results show that, in the tested IPv6 setting, external service exposure was determined primarily by the network perimeter rather than by the device. With the firmware, services, credentials and addressing behavior held constant, changing only the residential firewall state was sufficient to make the difference between services that were externally unreachable and services that were directly accessible from the Internet. IPv6 restores the architectural possibility of end-to-end connectivity, but whether that possibility becomes exposure depends on the filtering policy enforced at the residential boundary. The findings also show that accessibility and observability are not equivalent. Even when the firewall discarded all inbound TCP traffic and rendered the HTTP and Telnet services unreachable, the device continued to respond to ICMPv6 and remained identifiable through its EUI-64-derived address. A restrictive perimeter can therefore reduce service accessibility without eliminating discoverability or hardware-derived identification. Once the perimeter permits inbound traffic, however, the behavior of the exposed services becomes directly relevant. The same plaintext administrative access, default credentials, and undocumented endpoint that remained confined to the local network under restrictive perimeter conditions became reachable from outside the residential network without any modification to the device, while the IPv4/NAT baseline kept the same services unreachable without port forwarding. These observations describe the shift captured in the title of this thesis, from security by obscurity to imminent exposure. The protections that residential IoT deployments have often relied upon, specifically the isolation effect commonly provided by IPv4/NAT and the assumed difficulty of large-scale IPv6 host discovery, were never properties of the devices themselves but external conditions that the transition to IPv6 progressively removes. Within the scope of the conditions examined here, the factors that determined exposure were established primarily by residential filtering policies and device-side addressing behavior rather than by actions available to the end user. Reducing this class of risk across the residential ecosystem therefore depends largely on the actors that define those conditions, including ISPs, router manufacturers, and device manufacturers.

7.1 Future Work Several directions for future research follow from the scope and limitations of this work. The experiment evaluated reachability under known-address conditions, using the global IPv6 address obtained directly from the device. Extending the analysis to the discovery phase would help characterize the complete path from host identification to service access. This includes evaluating whether the same device could be located through adaptive target-generation techniques or inferred through ICMP-related side channels under realistic residential conditions. A second direction concerns the diversity of residential deployments. The observations reported here are based on a single router model evaluated under two perimeter conditions. Expanding the analysis to additional residential equipment, filtering policies, and deployment models would clarify how representative the observed behavior is across the broader ecosystem. This includes intermediate filtering configurations, ISP-defined defaults, alternative IPv6 transition mechanisms, and access architectures that differ from the native dual-stack environment examined in this work. The threat model examined here focused on the conditions that enabled external access rather than on the consequences that follow from it. Future work could therefore extend the analysis beyond successful authentication and service access to evaluate post-access activity under controlled conditions, including per-

sistence, lateral movement, and other forms of compromise. Finally, at the device level, additional research is also needed on the adoption of privacy-preserving addressing mechanisms in constrained IoT environments. Evaluating the feasibility, performance impact, and implementation cost of alternatives to EUI-64 in resource-limited devices would help address the identifiability dimension that perimeter controls cannot eliminate.

Bibliography

- [1] S. Sinha, *State of IoT 2024: Number of connected IoT devices growing 13% to 18.8 billion globally*, <https://iot-analytics.com/number-of-connected-iot-devices-2024/>, Accessed: 2026-02-23, Sep. 2024.
- [2] P. Srisuresh and K. B. Egevang, *Traditional IP network address translator (traditional NAT)*, RFC 3022, Jan. 2001. DOI: 10.17487/RFC3022 [Online]. Available: <https://www.rfc-editor.org/info/rfc3022>
- [3] F. Audet and C. Jennings, *Network address translation (NAT) behavioral requirements for unicast UDP*, RFC 4787, Jan. 2007. DOI: 10.17487/RFC4787 [Online]. Available: <https://www.rfc-editor.org/info/rfc4787>
- [4] J. H. Saltzer, D. P. Reed, and D. D. Clark, “End-to-end arguments in system design,” *ACM Transactions on Computer Systems*, vol. 2, no. 4, pp. 277–288, Nov. 1984. DOI: 10.1145/357401.357402
- [5] S. E. Deering and B. Hinden, *Internet protocol, version 6 (IPv6) specification*, RFC 8200, Jul. 2017. DOI: 10.17487/RFC8200 [Online]. Available: <https://www.rfc-editor.org/info/rfc8200>
- [6] E. Rye, D. Levin, and R. Beverly, “Where have all the firewalls gone? Security consequences of residential IPv6 transition,” *arXiv preprint arXiv:2509.04792*, Sep. 2025. [Online]. Available: <https://arxiv.org/abs/2509.04792>
- [7] M. Antonakakis et al., “Understanding the Mirai botnet,” in *Proceedings of the 26th USENIX Security Symposium*, Aug. 2017, pp. 1093–1110. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis>
- [8] A. Costin, J. Zaddach, A. Francillon, and D. Balzarotti, “A large-scale analysis of the security of embedded firmwares,” in *Proceedings of the 23rd USENIX Security Symposium*, Aug. 2014, pp. 95–110. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity14/technical-sessions/presentation/costin>
- [9] E. Fernandes, J. Jung, and A. Prakash, “Security analysis of emerging smart home applications,” in *Proceedings of the 2016 IEEE Symposium on Security and Privacy (S&P)*, May 2016, pp. 636–654. DOI: 10.1109/SP.2016.44 [Online]. Available: <https://doi.org/10.1109/SP.2016.44>
- [10] J. Czyz, M. Luckie, M. Allman, and M. Bailey, “Don’t forget to lock the back door! A characterization of IPv6 network security policy,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, Internet Society, Feb. 2016. DOI: 10.14722/ndss.2016.23047 [Online]. Available: https://faculty.cc.gatech.edu/~mbailey/publications/ndss16_ipv6.pdf
- [11] O. Alrawi, C. Lever, M. Antonakakis, and F. Monrose, “SoK: Security evaluation of home-based IoT deployments,” in *Proceedings of the 2019 IEEE Symposium on Security and Privacy (S&P)*, May 2019, pp. 1362–1380. DOI: 10.1109/SP.2019.00013 [Online]. Available: <https://www.computer.org/csdl/proceedings-article/sp/2019/666000b362/1dlwjLHaXwk>
- [12] H. Liu et al., “EAGLEYE: Exposing hidden web interfaces in IoT devices via routing analysis,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, Internet Society, Feb. 2025. DOI: 10.14722/ndss.2025.240399 [Online]. Available: <https://www.ndss-symposium.org/ndss-paper/eagleeye-exposing-hidden-web-interfaces-in-iot-devices-via-routing-analysis/>

- [13] Y. Shoshitaishvili, R. Wang, C. Hauser, C. Kruegel, and G. Vigna, “Firmalice: Automatic detection of authentication bypass vulnerabilities in binary firmware,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, Internet Society, Feb. 2015. DOI: 10.14722/ndss.2015.23294 [Online]. Available: <https://www.ndss-symposium.org/ndss2015/ndss-2015-programme/firmalice-automatic-detection-authentication-bypass-vulnerabilities-binary-firmware/>
- [14] G. Williams et al., “6Sense: Internet-Wide IPv6 scanning and its security applications,” in *Proceedings of the 33rd USENIX Security Symposium*, Aug. 2024, pp. 2281–2298. [Online]. Available: <https://www.usenix.org/system/files/usenixsecurity24-williams.pdf>
- [15] P. Nikander, J. Kempf, and E. Nordmark, *IPv6 neighbor discovery (ND) trust models and threats*, RFC 3756, May 2004. DOI: 10.17487/RFC3756 [Online]. Available: <https://www.rfc-editor.org/info/rfc3756>
- [16] E. C. Rye and R. Beverly, “IPvSeeYou: Exploiting leaked identifiers in IPv6 for street-level geolocation,” in *Proceedings of the 2023 IEEE Symposium on Security and Privacy (S&P)*, May 2023, pp. 1862–1880. DOI: 10.1109/SP46215.2023.10179376 [Online]. Available: <https://ieeexplore.ieee.org/document/10179376>
- [17] S. Thomson, T. Narten, and T. Jinmei, “IPv6 stateless address autoconfiguration,” Internet Engineering Task Force (IETF), Tech. Rep. RFC 4862, Sep. 2007. DOI: 10.17487/RFC4862 [Online]. Available: <https://www.rfc-editor.org/info/rfc4862>
- [18] T. Narten, E. Nordmark, W. A. Simpson, and H. Soliman, “Neighbor discovery for IP version 6 (IPv6),” Internet Engineering Task Force (IETF), Tech. Rep. RFC 4861, Sep. 2007. DOI: 10.17487/RFC4861 [Online]. Available: <https://www.rfc-editor.org/info/rfc4861>
- [19] J. Arkko, J. Kempf, B. Zill, and P. Nikander, *SEcure neighbor discovery (SEND)*, RFC 3971, Mar. 2005. DOI: 10.17487/RFC3971 [Online]. Available: <https://www.rfc-editor.org/info/rfc3971>
- [20] A. Conta, S. E. Deering, and M. Gupta, “Internet control message protocol (ICMPv6) for the internet protocol version 6 (IPv6) specification,” Internet Engineering Task Force (IETF), Tech. Rep. RFC 4443, Mar. 2006. DOI: 10.17487/RFC4443 [Online]. Available: <https://www.rfc-editor.org/info/rfc4443>
- [21] E. Davies and J. Mohácsi, *Recommendations for filtering ICMPv6 messages in firewalls*, RFC 4890, May 2007. DOI: 10.17487/RFC4890 [Online]. Available: <https://www.rfc-editor.org/info/rfc4890>
- [22] L. Yu, B. Luo, J. Ma, Z. Zhou, and Q. Liu, “You are what you broadcast: Identification of mobile and IoT devices from (public) WiFi,” in *Proceedings of the 29th USENIX Security Symposium*, Aug. 2020, pp. 55–72. [Online]. Available: <https://www.usenix.org/system/files/sec20-yu.pdf>
- [23] Z. Durumeric, E. Wustrow, and J. A. Halderman, “ZMap: Fast internet-wide scanning and its security applications,” in *Proceedings of the 22nd USENIX Security Symposium*, Aug. 2013, pp. 605–620. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity13/technical-sessions/paper/durumeric>
- [24] Z. Durumeric, D. Adrian, A. Mirian, M. Bailey, and J. A. Halderman, “A search engine backed by internet-wide scanning,” in *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security (CCS)*, Oct. 2015, pp. 542–553. DOI: 10.1145/2810103.2813703 [Online]. Available: <https://dl.acm.org/doi/10.1145/2810103.2813703>
- [25] L. Pan et al., “Your router is my prober: Measuring IPv6 networks via ICMP rate limiting side channels,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, Internet Society, Feb. 2023. DOI: 10.14722/ndss.2023.23049 [Online]. Available: <https://www.ndss-symposium.org/ndss-paper/your-router-is-my-prober-measuring-ipv6-networks-via-icmp-rate-limiting-side-channels/>

- [26] J. Woodyatt, "Recommended simple security capabilities in customer premises equipment (cpe) for providing residential ipv6 internet service," Internet Engineering Task Force (IETF), Tech. Rep. RFC 6092, Jan. 2011. DOI: 10.17487/RFC6092 [Online]. Available: <https://www.rfc-editor.org/info/rfc6092>
- [27] F. Gont, "A method for generating semantically opaque interface identifiers with IPv6 stateless address autoconfiguration (SLAAC)," Internet Engineering Task Force (IETF), Tech. Rep. RFC 7217, Apr. 2014. DOI: 10.17487/RFC7217 [Online]. Available: <https://www.rfc-editor.org/info/rfc7217>
- [28] F. Gont, S. Krishnan, T. Narten, and R. Draves, "Temporary address extensions for stateless address autoconfiguration in IPv6," Internet Engineering Task Force (IETF), Tech. Rep. RFC 8981, Feb. 2021. DOI: 10.17487/RFC8981 [Online]. Available: <https://www.rfc-editor.org/info/rfc8981>

A. Source Code

This appendix contains the complete source code for the ESP32 firmware used throughout the experiments. The firmware implements the network services (HTTP and Telnet) over IPv6 and manages the device's network identity.

```
#include <Arduino.h>
#include <WiFi.h>
#include "esp_event.h"
#include "esp_netif.h"
#include "lwip/ip6_addr.h"
#include "lwip/sockets.h"
#include "lwip/inet.h"

const char* ssid = "[REDACTED]";
const char* password = "[REDACTED]";

static esp_netif_t* sta_netif = nullptr;

esp_ip6_addr_t g_ll_addr;
esp_ip6_addr_t g_global_addr;
bool g_has_ll = false;
bool g_has_global = false;

// =====
// Counters
// =====
uint32_t http_hits_total = 0;
uint32_t hidden_admin_hits = 0;
uint32_t telnet_connections = 0;
uint32_t telnet_login_success = 0;
uint32_t telnet_login_failure = 0;

// =====
// Timestamp helper (ms since boot)
// =====
String ts() {
    return "[" + String(millis()) + "ms]";
}

// =====
// Helpers
// =====
String ipv6ToString(const esp_ip6_addr_t& addr) {
    char buf[40];
    ip6addr_ntoa_r((const ip6_addr_t*)&addr, buf, sizeof(buf));
```

```

    return String(buf);
}

String sockaddrIn6ToString(const sockaddr_in6* addr) {
    char buf[INET6_ADDRSTRLEN];
    inet_ntop(AF_INET6, &(addr->sin6_addr), buf, sizeof(buf));
    return String(buf);
}

void printIPv6(const esp_ip6_addr_t& addr) {
    Serial.println(ipv6ToString(addr));
}

String readLineFromSocket(int clientSock) {
    String line = "";
    char c;
    while (true) {
        int n = recv(clientSock, &c, 1, 0);
        if (n <= 0) break;
        if (c == '\r') continue;
        if (c == '\n') break;
        line += c;
        if (line.length() > 200) break;
    }
    return line;
}

void printCounters() {
    Serial.println("=== COUNTERS ===");
    Serial.println(" http_hits_total: " + String(http_hits_total));
    Serial.println(" hidden_admin_hits: " + String(hidden_admin_hits));
    Serial.println(" telnet_connections: " + String(telnet_connections));
    Serial.println(" telnet_login_success: " + String(telnet_login_success));
    Serial.println(" telnet_login_failure: " + String(telnet_login_failure));
    Serial.println("=====");
}

// =====
// IPv6 Event Handler
// =====
void onGotIPv6(void* arg, esp_event_base_t event_base,
               int32_t event_id, void* event_data) {
    if (event_base == IP_EVENT && event_id == IP_EVENT_GOT_IP6) {
        ip_event_got_ip6_t* event = (ip_event_got_ip6_t*)event_data;
        Serial.print(ts() + " [IPv6] Address acquired: ");
        printIPv6(event->ip6_info.ip);

        if (sta_netif != nullptr) {

```

```

        if (esp_netif_get_ip6_linklocal(sta_netif, &g_ll_addr) == ESP_OK)
            g_has_ll = true;
        if (esp_netif_get_ip6_global(sta_netif, &g_global_addr) == ESP_OK)
            g_has_global = true;
    }
}
}

// =====
// HTTP Server (port 80)
// =====
void httpServerTask(void* pvParameters) {
    int serverSock = socket(AF_INET6, SOCK_STREAM, IPPROTO_TCP);
    if (serverSock < 0) {
        Serial.println(ts() + " [HTTP] Failed to create IPv6 socket");
        vTaskDelete(NULL);
        return;
    }

    int opt = 1;
    setsockopt(serverSock, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    sockaddr_in6 serverAddr;
    memset(&serverAddr, 0, sizeof(serverAddr));
    serverAddr.sin6_family = AF_INET6;
    serverAddr.sin6_addr = in6addr_any;
    serverAddr.sin6_port = htons(80);

    if (bind(serverSock, (struct sockaddr*)&serverAddr, sizeof(serverAddr)) < 0) {
        Serial.println(ts() + " [HTTP] bind() failed");
        close(serverSock);
        vTaskDelete(NULL);
        return;
    }

    if (listen(serverSock, 5) < 0) {
        Serial.println(ts() + " [HTTP] listen() failed");
        close(serverSock);
        vTaskDelete(NULL);
        return;
    }

    Serial.println(ts() + " [HTTP] Listening on port 80 (IPv6)");

    while (true) {
        sockaddr_in6 clientAddr;
        socklen_t clientLen = sizeof(clientAddr);

```

```

int clientSock = accept(serverSock, (struct sockaddr*)&clientAddr, &clientLen);
if (clientSock < 0) { delay(10); continue; }

String remoteIP = sockaddrIn6ToString(&clientAddr);
uint16_t remotePort = ntohs(clientAddr.sin6_port);

char buffer[1024];
int len = recv(clientSock, buffer, sizeof(buffer) - 1, 0);

if (len > 0) {
    buffer[len] = '\0';
    String request = String(buffer);
    int lineEnd = request.indexOf("\r\n");
    String firstLine = (lineEnd > 0) ? request.substring(0, lineEnd) : request;

    String path = "/";
    if (firstLine.startsWith("GET ")) {
        int secondSpace = firstLine.indexOf(' ', 4);
        if (secondSpace > 4) path = firstLine.substring(4, secondSpace);
    }

    http_hits_total++;
    Serial.println(ts() + " [HTTP] src=" + remoteIP
        + " port=" + String(remotePort)
        + " path=" + path);

    String body;
    String status = "HTTP/1.1 200 OK\r\n";

    if (path == "/") {
        body = "<html><body>";
        body += "<h1>ESP32 IoT Device Dashboard</h1>";
        body += "<p>Status: Online</p>";
        body += "<p>Device: ESP32</p>";
        body += "<p>Service: IPv6 test node</p>";
        body += "<p>Firmware: 1.0-demo</p>";
        body += "<p>MAC: " + WiFi.macAddress() + "</p>";
        if (g_has_global) body += "<p>Global IPv6: " + ipv6ToString(g_global_addr) + "</p>";
        if (g_has_ll) body += "<p>Link-local: " + ipv6ToString(g_ll_addr) + "</p>";
        body += "<hr>";
        body += "<p>HTTP hits: " + String(http_hits_total) + "</p>";
        body += "<p>Hidden admin hits: " + String(hidden_admin_hits) + "</p>";
        body += "<p>Telnet connections: " + String(telnet_connections) + "</p>";
        body += "</body></html>";
    }
    else if (path == "/hidden_admin") {
        hidden_admin_hits++;
        Serial.println(ts() + " [HTTP] *** HIDDEN ADMIN accessed from " + remoteIP + " ***");
    }
}

```

```

        body = "<html><body>";
        body += "<h1>Hidden Admin Interface</h1>";
        body += "<p>Diagnostic Mode: ENABLED</p>";
        body += "<p>Debug Access: TRUE</p>";
        body += "<p>Warning: undocumented endpoint</p>";
        body += "<p>MAC: " + WiFi.macAddress() + "</p>";
        body += "</body></html>";
    }
    else {
        status = "HTTP/1.1 404 Not Found\r\n";
        body = "<html><body><h1>404 Not Found</h1></body></html>";
    }

    String response = status
        + "Content-Type: text/html\r\n"
        + "Content-Length: " + String(body.length()) + "\r\n"
        + "Connection: close\r\n\r\n"
        + body;

    send(clientSock, response.c_str(), response.length(), 0);
}

close(clientSock);
printCounters();
}
}

// =====
// Telnet-like TCP Service (port 23)
// =====
void telnetServerTask(void* pvParameters) {
    int serverSock = socket(AF_INET6, SOCK_STREAM, IPPROTO_TCP);
    if (serverSock < 0) {
        Serial.println(ts() + " [TELNET] Failed to create IPv6 socket");
        vTaskDelete(NULL);
        return;
    }

    int opt = 1;
    setsockopt(serverSock, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    sockaddr_in6 serverAddr;
    memset(&serverAddr, 0, sizeof(serverAddr));
    serverAddr.sin6_family = AF_INET6;
    serverAddr.sin6_addr = in6addr_any;
    serverAddr.sin6_port = htons(23);

    if (bind(serverSock, (struct sockaddr*)&serverAddr, sizeof(serverAddr)) < 0) {

```

```

Serial.println(ts() + " [TELNET] bind() failed");
close(serverSock);
vTaskDelete(NULL);
return;
}

if (listen(serverSock, 2) < 0) {
    Serial.println(ts() + " [TELNET] listen() failed");
    close(serverSock);
    vTaskDelete(NULL);
    return;
}

Serial.println(ts() + " [TELNET] Listening on port 23 (IPv6)");

while (true) {
    sockaddr_in6 clientAddr;
    socklen_t clientLen = sizeof(clientAddr);

    int clientSock = accept(serverSock, (struct sockaddr*)&clientAddr, &clientLen);
    if (clientSock < 0) { delay(10); continue; }

    telnet_connections++;
    String remoteIP = sockaddrIn6ToString(&clientAddr);
    uint16_t remotePort = ntohs(clientAddr.sin6_port);

    Serial.println(ts() + " [TELNET] src=" + remoteIP
        + " port=" + String(remotePort) + " connected");

    String banner = "ESP32 Legacy Telnet Service\r\nUsername: ";
    send(clientSock, banner.c_str(), banner.length(), 0);

    String user = readLineFromSocket(clientSock);
    Serial.println(ts() + " [TELNET] src=" + remoteIP + " username=" + user);

    String askPass = "Password: ";
    send(clientSock, askPass.c_str(), askPass.length(), 0);

    String pass = readLineFromSocket(clientSock);
    Serial.println(ts() + " [TELNET] src=" + remoteIP + " password=" + pass);

    if (user == "admin" && pass == "admin") {
        telnet_login_success++;
        Serial.println(ts() + " [TELNET] src=" + remoteIP + " login=SUCCESS");

        String ok = "\r\nAccess granted\r\nType 'status' or 'exit'\r\n> ";
        send(clientSock, ok.c_str(), ok.length(), 0);
    }
}

```

```

while (true) {
    String cmd = readLineFromSocket(clientSock);
    if (cmd.length() == 0) break;

    Serial.println(ts() + " [TELNET] src=" + remoteIP + " cmd=" + cmd);

    if (cmd == "exit") {
        String bye = "Bye\r\n";
        send(clientSock, bye.c_str(), bye.length(), 0);
        break;
    } else if (cmd == "status") {
        String msg = "Device status: ONLINE\r\n";
        msg += "MAC: " + WiFi.macAddress() + "\r\n";
        if (g_has_global) msg += "Global IPv6: " + ipv6ToString(g_global_addr) + "\r\n";
        msg += "HTTP hits: " + String(http_hits_total) + "\r\n";
        msg += "Telnet sessions: " + String(telnet_connections) + "\r\n";
        msg += "Login successes: " + String(telnet_login_success) + "\r\n";
        msg += "> ";
        send(clientSock, msg.c_str(), msg.length(), 0);
    } else {
        String msg = "Unknown command\r\n> ";
        send(clientSock, msg.c_str(), msg.length(), 0);
    }
}

telnet_login_failure++;
Serial.println(ts() + " [TELNET] src=" + remoteIP + " login=FAILED");
String denied = "\r\nAccess denied\r\n";
send(clientSock, denied.c_str(), denied.length(), 0);
}

close(clientSock);
printCounters();
}
}

// =====
// Setup
// =====
void setup() {
    Serial.begin(115200);
    delay(1000);

    WiFi.mode(WIFI_STA);
    WiFi.setSleep(false); // Keep WiFi active prevents disconnection during experiments

    esp_event_handler_register(IP_EVENT, IP_EVENT_GOT_IP6, &onGotIPv6, nullptr);

```

```

Serial.println("=== ESP32 IPv6 Thesis Demo ===");
Serial.println("Connecting to WiFi...");
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Waiting for WiFi...");
}

Serial.println("WiFi connected");
Serial.print("IPv4: "); Serial.println(WiFi.localIP());
Serial.print("MAC: "); Serial.println(WiFi.macAddress());

sta_netif = esp_netif_get_handle_from_ifkey("WIFI_STA_DEF");
if (sta_netif == nullptr) {
    Serial.println("[ERROR] Could not get WIFI_STA_DEF handle");
    return;
}

esp_err_t err = esp_netif_create_ip6_linklocal(sta_netif);
Serial.print("Link-local IPv6 init: ");
Serial.println(err == ESP_OK ? "OK" : "FAILED");

delay(3000); // Wait for SLAAC global address assignment

xTaskCreatePinnedToCore(httpServerTask, "httpTask", 8192, NULL, 1, NULL, 1);
xTaskCreatePinnedToCore(telnetServerTask, "telnetTask", 8192, NULL, 1, NULL, 1);
}

// =====
// Loop prints identity every 100s
// =====
void loop() {
    if (sta_netif != nullptr) {
        esp_ip6_addr_t ll_addr, global_addr;
        esp_err_t ll_ok = esp_netif_get_ip6_linklocal(sta_netif, &ll_addr);
        esp_err_t gl_ok = esp_netif_get_ip6_global(sta_netif, &global_addr);

        Serial.println("=== DEVICE IDENTITY ===");
        Serial.print("MAC: "); Serial.println(WiFi.macAddress());
        Serial.print("IPv4: "); Serial.println(WiFi.localIP());

        if (ll_ok == ESP_OK) {
            g_ll_addr = ll_addr; g_has_ll = true;
            Serial.print("Link-local: "); printIPv6(ll_addr);
        } else {
            Serial.println("Link-local: not yet available");
        }
    }
}

```

```
    if (gl_ok == ESP_OK) {
        g_global_addr = global_addr; g_has_global = true;
        Serial.print("Global IPv6: "); printIPv6(global_addr);
    } else {
        Serial.println("Global IPv6: not yet available");
    }

    Serial.println("=====");
    printCounters();
}

delay(100000);
}
```

Listing A.1: ESP32 Firmware Source Code (main.cpp)

B. Experimental Results

The following sections document the raw evidence collected during the execution of each experimental scenario. This data includes terminal outputs, network probes, and device-level logs.

B.1 Scenario 0: Baseline (Internal Network) The objective of this scenario is to verify the internal availability of all services on the LAN before any external reachability tests.

B.1.1 Local Network Discovery and ICMPv6 The following logs show the device identifying its IPv6 global and link-local addresses.

```
=== ESP32 IPv6 Thesis Demo ===
Connecting to WiFi...
Waiting for WiFi...
WiFi connected
IPv4: 192.168.0.10
MAC: 28:05:A5:0E:73:6C
Link-local IPv6 init: OK
[4028ms] [IPv6] Address acquired: FE80::2A05:A5FF:FE0E:736C
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: not yet available
[5137ms] [TELNET] Listening on port 23 (IPv6)
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
[5137ms] [HTTP] Listening on port 80 (IPv6)
[8028ms] [IPv6] Address acquired: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
```

```

telnet_login_success: 0
telnet_login_failure: 0
=====
                                                                28:05:A5:0E:73:6C

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 0
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
                                                                9AE1:B100:910E:1492:DE64:3BBD port=18403 path=/

=== COUNTERS ===
  http_hits_total: 1
  hidden_admin_hits: 0
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
[274167ms] [HTTP] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=45657 path=/hidden_admin
[274167ms] [HTTP] *** HIDDEN ADMIN accessed from 2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD ***
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 0
  telnet_login_success: 0
  telnet_login_failure: 0
=====
                                                                28:05:A5:0E:73:6C

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C

```

```

Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 0
    telnet_login_success: 0
    telnet_login_failure: 0
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 0
    telnet_login_success: 0
    telnet_login_failure: 0
=====
[593275ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=4575 connected
[600817ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username= ' admin
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 1
    telnet_login_success: 0
    telnet_login_failure: 0
=====
[610552ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=admin
[610553ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=FAILED
=== COUNTERS ===
    http_hits_total: 2
    hidden_admin_hits: 1
    telnet_connections: 1
    telnet_login_success: 0
    telnet_login_failure: 1
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C

```

```

IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 1
  telnet_login_success: 0
  telnet_login_failure: 1
=====
[727179ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=17466 connected
[738311ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username=
[740739ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=
[740740ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=FAILED
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 2
  telnet_login_success: 0
  telnet_login_failure: 2
=====
[752131ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD port=17521 connected
[755063ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD username=admin
[757505ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD password=admin
[757505ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD login=SUCCESS
[762498ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD cmd=status
[767211ms] [TELNET] src=2A02:21B4:9AE1:B100:910E:1492:DE64:3BBD cmd=exit
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 3
  telnet_login_success: 1
  telnet_login_failure: 2
=====
=== DEVICE IDENTITY ===
MAC: 28:05:A5:0E:73:6C
IPv4: 192.168.0.10
Link-local: FE80::2A05:A5FF:FE0E:736C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
=====
=== COUNTERS ===
  http_hits_total: 2
  hidden_admin_hits: 1
  telnet_connections: 3
  telnet_login_success: 1
  telnet_login_failure: 2
=====

```

Listing B.1: S0 - Device Serial Log

B.1.2 Internal Service Verification

```

StatusCode : 200
StatusDescription : OK
Content : <html><body><h1>ESP32 IoT Device Dashboard</h1><p>Status: Online</p><p>Device:
          ESP32</p><p>Service: IPv6 test node</p><p>Firmware: 1.0-demo</p><p>MAC:
          28:05:A5:0E:73:6C</p><p>Global IPv6: 2A02:21B4:9A...
RawContent : HTTP/1.1 200 OK
              Connection: close
              Content-Length: 367
              Content-Type: text/html

              <html><body><h1>ESP32 IoT Device Dashboard</h1><p>Status: Online</p><p>Device:
              ESP32</p><p>Service: IPv6 test node</p>...
Forms : {}
Headers : {[Connection, close], [Content-Length, 367], [Content-Type, text/html]}
Images : {}
InputFields : {}
Links : {}
ParsedHtml : mshtml.HTMLDocumentClass
RawContentLength : 367
```

Listing B.2: S0 - Local HTTP Root Request

```

StatusCode : 200
StatusDescription : OK
Content : <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access:
          TRUE</p><p>Warning: undocumented endpoint</p><p>MAC: 28:05:A5:0E:73:6C</p></body></ht
RawContent : HTTP/1.1 200 OK
              Connection: close
              Content-Length: 179
              Content-Type: text/html

              <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Ac
              TRUE</p><p>Warning: undoc...
Forms : {}
Headers : {[Connection, close], [Content-Length, 179], [Content-Type, text/html]}
Images : {}
InputFields : {}
Links : {}
ParsedHtml : mshtml.HTMLDocumentClass
RawContentLength : 179
```

Listing B.3: S0 - Local HTTP Hidden Admin Request

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 16:53 +0100
Nmap scan report for esp32-0e736c.home (2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c)
Host is up (0.0084s latency).

PORT STATE SERVICE
23/tcp open telnet
80/tcp open http

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

Listing B.4: S0 - Local Nmap Scan Output

B.2 Scenario 1: External Reachability (Firewall ON) In this scenario, an external VPS attempts to reach the device while the residential router's IPv6 firewall is active.

B.2.1 Remote Reachability and Service Probing

```
PING 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C (2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c) 56 data bytes
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=1 ttl=238 time=41.9 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=2 ttl=238 time=44.1 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=3 ttl=238 time=38.6 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=4 ttl=238 time=39.1 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=5 ttl=238 time=38.8 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=6 ttl=238 time=41.7 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=7 ttl=238 time=38.7 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=8 ttl=238 time=43.8 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=9 ttl=238 time=41.0 ms
64 bytes from 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c: icmp_seq=10 ttl=238 time=40.1 ms

--- 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 38.627/40.787/44.108/1.964 ms
```

Listing B.5: S1 - External ICMPv6 Ping

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:12 UTC
Nmap scan report for 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Host is up, received user-set.

PORT STATE SERVICE REASON
23/tcp filtered telnet no-response
80/tcp filtered http no-response

Nmap done: 1 IP address (1 host up) scanned in 3.46 seconds
```

Listing B.6: S1 - External Nmap Scan (Firewall Active)

B.3 Scenario 2: Service Exposure (Firewall OFF) This scenario demonstrates the full exposure of the IoT services when the IPv6 firewall is disabled or misconfigured.

B.3.1 External Service Access

```
ssh : % Total % Received % Xferd Average Speed Time Time Time
      Current
At line:1 char:1
+ ssh $VPS "curl -v --max-time 10 http://[$ESP32]/hidden_admin" 2>&1 ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: ( % Total % ... Time Curre
nt:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

Dload Upload Total Spent Left Speed

 0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:--
0* Trying [2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c]:80...
* Connected to 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
(2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c) port 80
> GET /hidden_admin HTTP/1.1
> Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
< Content-Length: 179
< Connection: close
<
{ [179 bytes data]

100 179 100 179 0 0 1757 0 --:--:-- --:--:-- --:--:--
1772
* Closing connection
<html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p><p>W
```

Listing B.7: S2 - External HTTP Hidden Admin Access

```
Trying 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c...
Connected to 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C.
Escape character is '^]'.
ESP32 Legacy Telnet Service
Username: Password:
Access granted
Type 'status' or 'exit'
> Device status: ONLINE
MAC: 28:05:A5:0E:73:6C
Global IPv6: 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C
HTTP hits: 2
```

```
Telnet sessions: 2
Login successes: 1
> Bye
```

Listing B.8: S2 - External Telnet Session Attempt

B.3.2 Packet Analysis and Remote Discovery

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:18 UTC
Nmap scan report for 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Host is up, received user-set (0.045s latency).

PORT STATE SERVICE REASON
23/tcp open  telnet syn-ack
80/tcp open  http  syn-ack

Nmap done: 1 IP address (1 host up) scanned in 0.26 seconds
```

Listing B.9: S2 - Remote Port Scan Result

```
root@thesis:~# sudo tcpdump -i any -nn -A -c 20 "ip6 and host 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C and tcp port
sleep 1 && curl -v "http://[2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]/hidden_admin"
wait
[1] 7925
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
* Trying [2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c]:80...
16:24:29.728803 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80: Flags [S], s
'....(.*....\.....*!.....*.....sl...P.{..... $t.....
...%.
* Connected to 2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C (2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c) port 80
> GET /hidden_admin HTTP/1.1
> Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
< Content-Length: 179
< Connection: close
<
* Closing connection
<html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p><p>Warning: undoc
In IP6 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80 > 2a01:4f9:c014:5c1d::1.45744: Flags [S.], seq 3950055805, ack 21
'.....0*!.....*.....sl*.....\.....P...q.}.{..'.f.....
16:24:29.783781 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80: Flags [.], a
'.....@*....\.....*!.....*.....sl...P.{...q.~P.. $'..
16:24:29.783943 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80: Flags [P.],
'.....@*....\.....*!.....*.....sl...P.{...q.~P.. $...GET /hidden_admin HTTP/1.1
```

```
Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]
User-Agent: curl/8.5.0
Accept: */*

16:24:29.837630 eth0 In IP6 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80 > 2a01:4f9:c014:5c1d::1.45744: F
'.....0*!.!.....*.....sl*.....\.....P...q...{..P....d..HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 179
Connection: close

<html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p><p>W
16:24:29.837688 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80:
'.....@*.....\.....*!.!.....*.....sl...P.{...q..P...$'..
16:24:29.837858 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80:
'.....@*.....\.....*!.!.....*.....sl...P.{...q..P...$'..
16:24:29.839556 eth0 In IP6 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80 > 2a01:4f9:c014:5c1d::1.45744: F
'.....0*!.!.....*.....sl*.....\.....P...q...{..P...}~..
16:24:29.839570 eth0 Out IP6 2a01:4f9:c014:5c1d::1.45744 > 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80:
'.....@*.....\.....*!.!.....*.....sl...P.{...q..P...$'..
16:24:29.891632 eth0 In IP6 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c.80 > 2a01:4f9:c014:5c1d::1.45744: F
'.....0*!.!.....*.....sl*.....\.....P...q...{..P...}~..
```

Listing B.10: S2 - External Packet Capture (Plaintext HTTP Request/Response)

B.4 Scenario 3: Local Traffic Observation This scenario validates that sensitive information remains exposed in plaintext to local observers within the LAN.

B.4.1 Deep Packet Inspection (DPI) The following captures demonstrate that HTTP and Telnet payloads are visible within the network segment.

```
No. Time Source Destination Protocol Length Info
62433 123.312874 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
86 6250 -> 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

Frame 62433: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.458268000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.458268000 UTC
Epoch Arrival Time: 1774456274.458268000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 6.246000 milliseconds]
[Time since reference or first frame: 2 minutes, 3.312874000 seconds]
Frame Number: 62433
Frame Length: 86 bytes (688 bits)
Capture Length: 86 bytes (688 bits)
[Frame is marked: False]
```

```

[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 0, Len: 0

No. Time Source Destination Protocol Length Info
62586 123.634538 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
78 80 -> 6250 [SYN, ACK] Seq=0 Ack=1 Win=5760 Len=0 MSS=1412

Frame 62586: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.779932000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.779932000 UTC
Epoch Arrival Time: 1774456274.779932000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 7.760000 milliseconds]
[Time delta from previous displayed frame: 321.664000 milliseconds]
[Time since reference or first frame: 2 minutes, 3.634538000 seconds]
Frame Number: 62586
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 0, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
62587 123.634614 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6250 -> 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 62587: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.780008000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.780008000 UTC
Epoch Arrival Time: 1774456274.780008000

```

```

[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 76.000 microseconds]
[Time delta from previous displayed frame: 76.000 microseconds]
[Time since reference or first frame: 2 minutes, 3.634614000 seconds]
Frame Number: 62587
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
62588 123.635123 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c HTTP
272 GET /hidden_admin HTTP/1.1

Frame 62588: Packet, 272 bytes on wire (2176 bits), 272 bytes captured (2176 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:14.780517000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:14.780517000 UTC
Epoch Arrival Time: 1774456274.780517000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 509.000 microseconds]
[Time delta from previous displayed frame: 509.000 microseconds]
[Time since reference or first frame: 2 minutes, 3.635123000 seconds]
Frame Number: 62588
Frame Length: 272 bytes (2176 bits)
Capture Length: 272 bytes (2176 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:http]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 1, Ack: 1, Len: 198
Hypertext Transfer Protocol
GET /hidden_admin HTTP/1.1\r\n
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) WindowsPowerShell/5.1.26100.7920\r\n
Host: [2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]\r\n

```

```

Connection: Keep-Alive\r\n
\r\n
[Response in frame: 62595]
[Full request URI: http://[2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]/hidden_admin]

No. Time Source Destination Protocol Length Info
 62595 123.656565 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd HTTP
 337 HTTP/1.1 200 OK (text/html)

Frame 62595: Packet, 337 bytes on wire (2696 bits), 337 bytes captured (2696 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:31:14.801959000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:31:14.801959000 UTC
  Epoch Arrival Time: 1774456274.801959000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 16.340000 milliseconds]
  [Time delta from previous displayed frame: 21.442000 milliseconds]
  [Time since reference or first frame: 2 minutes, 3.656565000 seconds]
  Frame Number: 62595
  Frame Length: 337 bytes (2696 bits)
  Capture Length: 337 bytes (2696 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:http:data-text-lines]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 1, Ack: 199, Len: 263
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
  Content-Type: text/html\r\n
  Content-Length: 179\r\n
  Connection: close\r\n
  \r\n
  [Request in frame: 62588]
  [Time since request: 21.442000 milliseconds]
  [Request URI: /hidden_admin]
  [Full request URI: http://[2A02:21B4:9AE1:B100:2A05:A5FF:FE0E:736C]/hidden_admin]
  File Data: 179 bytes
Line-based text data: text/html (1 lines)
  <html><body><h1>Hidden Admin Interface</h1><p>Diagnostic Mode: ENABLED</p><p>Debug Access: TRUE</p><p>Warning:
No. Time Source Destination Protocol Length Info
 62597 123.660383 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP

```

74 80 -> 6250 [FIN, ACK] Seq=264 Ack=199 Win=5562 Len=0

Frame 62597: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:14.805777000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:14.805777000 UTC

Epoch Arrival Time: 1774456274.805777000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 94.000 microseconds]

[Time delta from previous displayed frame: 3.818000 milliseconds]

[Time since reference or first frame: 2 minutes, 3.660383000 seconds]

Frame Number: 62597

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: HTTP]

[Coloring Rule String: http tcp.port == 80 http2]

Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:9100:2a05:a5ff:fe0e:736c

Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 264, Ack: 199, Len: 0

No. Time Source Destination Protocol Length Info

62598 123.660416 2a02:21b4:9ae1:b100:9100:2a05:a5ff:fe0e:736c TCP

74 6250 -> 80 [ACK] Seq=199 Ack=265 Win=65272 Len=0

Frame 62598: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:14.805810000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:14.805810000 UTC

Epoch Arrival Time: 1774456274.805810000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 33.000 microseconds]

[Time delta from previous displayed frame: 33.000 microseconds]

[Time since reference or first frame: 2 minutes, 3.660416000 seconds]

Frame Number: 62598

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp]

Character encoding: ASCII (0)

<pre> [Coloring Rule Name: HTTP] [Coloring Rule String: http tcp.port == 80 http2] Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c) Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 199, Ack: 265, Len: 0 No. Time Source Destination Protocol Length Info 64162 126.585140 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP 74 6250 -> 80 [FIN, ACK] Seq=199 Ack=265 Win=65272 Len=0 Frame 64162: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8} Section number: 1 Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}) Encapsulation type: Ethernet (1) Arrival Time: Mar 25, 2026 17:31:17.730534000 W. Europe Standard Time UTC Arrival Time: Mar 25, 2026 16:31:17.730534000 UTC Epoch Arrival Time: 1774456277.730534000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 4.733000 milliseconds] [Time delta from previous displayed frame: 2.924724000 seconds] [Time since reference or first frame: 2 minutes, 6.585140000 seconds] Frame Number: 64162 Frame Length: 74 bytes (592 bits) Capture Length: 74 bytes (592 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: eth:ethertype:ipv6:tcp] Character encoding: ASCII (0) [Coloring Rule Name: HTTP] [Coloring Rule String: http tcp.port == 80 http2] Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c) Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c Transmission Control Protocol, Src Port: 6250, Dst Port: 80, Seq: 199, Ack: 265, Len: 0 </pre>	
<pre> No. Time Source Destination Protocol Length Info 64163 126.593359 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP 74 80 -> 6250 [ACK] Seq=265 Ack=200 Win=5561 Len=0 Frame 64163: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8} Section number: 1 Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}) Encapsulation type: Ethernet (1) Arrival Time: Mar 25, 2026 17:31:17.738753000 W. Europe Standard Time UTC Arrival Time: Mar 25, 2026 16:31:17.738753000 UTC Epoch Arrival Time: 1774456277.738753000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 8.219000 milliseconds] [Time delta from previous displayed frame: 8.219000 milliseconds] </pre>	


```

[Time since reference or first frame: 2 minutes, 6.593359000 seconds]
Frame Number: 64163
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http tcp.port == 80 http2]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 80, Dst Port: 6250, Seq: 265, Ack: 200, Len: 0

No. Time Source Destination Protocol Length Info
85498 166.512013 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
86 6336 -> 23 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

Frame 85498: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:57.657407000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:57.657407000 UTC
Epoch Arrival Time: 1774456317.657407000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.383000 milliseconds]
[Time delta from previous displayed frame: 39.918654000 seconds]
[Time since reference or first frame: 2 minutes, 46.512013000 seconds]
Frame Number: 85498
Frame Length: 86 bytes (688 bits)
Capture Length: 86 bytes (688 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 tcp.flags.fin == 1]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 0, Len: 0

No. Time Source Destination Protocol Length Info
85499 166.520588 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TCP
78 23 -> 6336 [SYN, ACK] Seq=0 Ack=1 Win=5760 Len=0 MSS=1412

Frame 85499: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1

```

```

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:57.665982000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:57.665982000 UTC
Epoch Arrival Time: 1774456317.665982000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 8.575000 milliseconds]
[Time delta from previous displayed frame: 8.575000 milliseconds]
[Time since reference or first frame: 2 minutes, 46.520588000 seconds]
Frame Number: 85499
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 tcp.flags.fin == 1]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 0, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info
85500 166.520657 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 85500: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:31:57.666051000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:31:57.666051000 UTC
Epoch Arrival Time: 1774456317.666051000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 69.000 microseconds]
[Time delta from previous displayed frame: 69.000 microseconds]
[Time since reference or first frame: 2 minutes, 46.520657000 seconds]
Frame Number: 85500
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c

```

Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 1, Len: 0

No. Time Source Destination Protocol Length Info

85503 166.530566 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET
113 39 bytes data

Frame 85503: Packet, 113 bytes on wire (904 bits), 113 bytes captured (904 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:57.675960000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:57.675960000 UTC

Epoch Arrival Time: 1774456317.675960000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 5.904000 milliseconds]

[Time delta from previous displayed frame: 9.909000 milliseconds]

[Time since reference or first frame: 2 minutes, 46.530566000 seconds]

Frame Number: 85503

Frame Length: 113 bytes (904 bits)

Capture Length: 113 bytes (904 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]

Character encoding: ASCII (0)

[Coloring Rule Name: TCP]

[Coloring Rule String: tcp]

Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd

Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 1, Ack: 1, Len: 39

Telnet

No. Time Source Destination Protocol Length Info

85536 166.580399 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=1 Ack=40 Win=65496 Len=0

Frame 85536: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:31:57.725793000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:31:57.725793000 UTC

Epoch Arrival Time: 1774456317.725793000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 182.000 microseconds]

[Time delta from previous displayed frame: 49.833000 milliseconds]

[Time since reference or first frame: 2 minutes, 46.580399000 seconds]

Frame Number: 85536

Frame Length: 74 bytes (592 bits)

```

Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 40, Len: 0

No. Time Source Destination Protocol Length Info
  86964 169.351640 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET
  81 7 bytes data

Frame 86964: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on interface \Device\NPF_{BE969710-4
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 25, 2026 17:32:00.497034000 W. Europe Standard Time
  UTC Arrival Time: Mar 25, 2026 16:32:00.497034000 UTC
  Epoch Arrival Time: 1774456320.497034000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 6.060000 milliseconds]
  [Time delta from previous displayed frame: 2.771241000 seconds]
  [Time since reference or first frame: 2 minutes, 49.351640000 seconds]
  Frame Number: 86964
  Frame Length: 81 bytes (648 bits)
  Capture Length: 81 bytes (648 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
  Character encoding: ASCII (0)
  [Coloring Rule Name: TCP]
  [Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 1, Ack: 40, Len: 7
Telnet

No. Time Source Destination Protocol Length Info
  86966 169.361376 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET
  84 10 bytes data

Frame 86966: Packet, 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface \Device\NPF_{BE969710-4
  Section number: 1
  Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
  Encapsulation type: Ethernet (1)

```

```

Arrival Time: Mar 25, 2026 17:32:00.506770000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:32:00.506770000 UTC
Epoch Arrival Time: 1774456320.506770000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.021000 milliseconds]
[Time delta from previous displayed frame: 9.736000 milliseconds]
[Time since reference or first frame: 2 minutes, 49.361376000 seconds]
Frame Number: 86966
Frame Length: 84 bytes (672 bits)
Capture Length: 84 bytes (672 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 40, Ack: 8, Len: 10
Telnet

No. Time Source Destination Protocol Length Info
86992 169.411843 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=8 Ack=50 Win=65486 Len=0

Frame 86992: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:32:00.557237000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:32:00.557237000 UTC
Epoch Arrival Time: 1774456320.557237000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 9.385000 milliseconds]
[Time delta from previous displayed frame: 50.467000 milliseconds]
[Time since reference or first frame: 2 minutes, 49.411843000 seconds]
Frame Number: 86992
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 8, Ack: 50, Len: 0

```

No. Time Source Destination Protocol Length Info

87954 171.420833 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET
81 7 bytes data

Frame 87954: Packet, 81 bytes on wire (648 bits), 81 bytes captured (648 bits) on **interface** \Device\NPF_{BE969710-4

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:32:02.566227000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:32:02.566227000 UTC

Epoch Arrival Time: 1774456322.566227000

[Time shift **for this** packet: 0.000000000 seconds]

[Time delta from previous captured frame: 741.000 microseconds]

[Time delta from previous displayed frame: 2.008990000 seconds]

[Time since reference or first frame: 2 minutes, 51.420833000 seconds]

Frame Number: 87954

Frame Length: 81 bytes (648 bits)

Capture Length: 81 bytes (648 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]

Character encoding: ASCII (0)

[Coloring Rule Name: TCP]

[Coloring Rule String: tcp]

Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:

Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 8, Ack: 50, Len: 7

Telnet

No. Time Source Destination Protocol Length Info

87962 171.439640 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET
119 45 bytes data

Frame 87962: Packet, 119 bytes on wire (952 bits), 119 bytes captured (952 bits) on **interface** \Device\NPF_{BE969710-

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:32:02.585034000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:32:02.585034000 UTC

Epoch Arrival Time: 1774456322.585034000

[Time shift **for this** packet: 0.000000000 seconds]

[Time delta from previous captured frame: 3.458000 milliseconds]

[Time delta from previous displayed frame: 18.807000 milliseconds]

[Time since reference or first frame: 2 minutes, 51.439640000 seconds]

Frame Number: 87962

Frame Length: 119 bytes (952 bits)

Capture Length: 119 bytes (952 bits)

```

[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 50, Ack: 15, Len: 45
Telnet

No. Time Source Destination Protocol Length Info
87977 171.482457 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=15 Ack=95 Win=65441 Len=0

Frame 87977: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:32:02.627851000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:32:02.627851000 UTC
Epoch Arrival Time: 1774456322.627851000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.080000 milliseconds]
[Time delta from previous displayed frame: 42.817000 milliseconds]
[Time since reference or first frame: 2 minutes, 51.482457000 seconds]
Frame Number: 87977
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 15, Ack: 95, Len: 0

No. Time Source Destination Protocol Length Info
92324 179.380451 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TELNET
82 8 bytes data

Frame 92324: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:32:10.525845000 W. Europe Standard Time

```

```

UTC Arrival Time: Mar 25, 2026 16:32:10.525845000 UTC
Epoch Arrival Time: 1774456330.525845000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 183.000 microseconds]
[Time delta from previous displayed frame: 7.897994000 seconds]
[Time since reference or first frame: 2 minutes, 59.380451000 seconds]
Frame Number: 92324
Frame Length: 82 bytes (656 bits)
Capture Length: 82 bytes (656 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 15, Ack: 95, Len: 8
Telnet

No. Time Source Destination Protocol Length Info
  92325 179.391380 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd TELNET
231 157 bytes data

Frame 92325: Packet, 231 bytes on wire (1848 bits), 231 bytes captured (1848 bits) on interface \Device\NPF_{BE9697
Section number: 1
Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 17:32:10.536774000 W. Europe Standard Time
UTC Arrival Time: Mar 25, 2026 16:32:10.536774000 UTC
Epoch Arrival Time: 1774456330.536774000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 10.929000 milliseconds]
[Time delta from previous displayed frame: 10.929000 milliseconds]
[Time since reference or first frame: 2 minutes, 59.391380000 seconds]
Frame Number: 92325
Frame Length: 231 bytes (1848 bits)
Capture Length: 231 bytes (1848 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp:telnet]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Espressif_0e:73:6c (28:05:a5:0e:73:6c), Dst: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c, Dst: 2a02:21b4:9ae1:b100:910e:1492:de64:
Transmission Control Protocol, Src Port: 23, Dst Port: 6336, Seq: 95, Ack: 23, Len: 157
Telnet

```


No. Time Source Destination Protocol Length Info

92344 179.444809 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [ACK] Seq=23 Ack=252 Win=65284 Len=0

Frame 92344: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:32:10.590203000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:32:10.590203000 UTC

Epoch Arrival Time: 1774456330.590203000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 4.653000 milliseconds]

[Time delta from previous displayed frame: 53.429000 milliseconds]

[Time since reference or first frame: 2 minutes, 59.444809000 seconds]

Frame Number: 92344

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ipv6:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: TCP]

[Coloring Rule String: tcp]

Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)

Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c

Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 23, Ack: 252, Len: 0

No. Time Source Destination Protocol Length Info

163041 299.395889 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:736c TCP
74 6336 -> 23 [RST, ACK] Seq=23 Ack=252 Win=0 Len=0

Frame 163041: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8}

Section number: 1

Interface id: 0 (\Device\NPF_{BE969710-4083-4F49-A1F4-592492C244C8})

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 17:34:10.541283000 W. Europe Standard Time

UTC Arrival Time: Mar 25, 2026 16:34:10.541283000 UTC

Epoch Arrival Time: 1774456450.541283000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 12.566000 milliseconds]

[Time delta from previous displayed frame: 1 minute, 59.951080000 seconds]

[Time since reference or first frame: 4 minutes, 59.395889000 seconds]

Frame Number: 163041

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

```
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP RST]
[Coloring Rule String: tcp.flags.reset eq 1]
Ethernet II, Src: AzureWaveTec_b3:4c:c3 (e8:fb:1c:b3:4c:c3), Dst: Espressif_0e:73:6c (28:05:a5:0e:73:6c)
Internet Protocol Version 6, Src: 2a02:21b4:9ae1:b100:910e:1492:de64:3bbd, Dst: 2a02:21b4:9ae1:b100:2a05:a5ff:fe0e:
Transmission Control Protocol, Src Port: 6336, Dst Port: 23, Seq: 23, Ack: 252, Len: 0
```

Listing B.11: S3 - Wireshark Terminal Export (Plaintext Traffic)

B.5 Scenario 4: Comparative Baseline (IPv4 / NAT) The final scenario provides a comparison with a traditional IPv4/NAT environment, where external reachability is denied by default.

B.5.1 IPv4 External Reachability Attempt

```
PING 198.51.100.7 (198.51.100.7) 56(84) bytes of data.
64 bytes from 198.51.100.7: icmp_seq=1 ttl=49 time=39.1 ms
64 bytes from 198.51.100.7: icmp_seq=2 ttl=49 time=34.8 ms
64 bytes from 198.51.100.7: icmp_seq=3 ttl=49 time=34.8 ms
64 bytes from 198.51.100.7: icmp_seq=4 ttl=49 time=34.7 ms
64 bytes from 198.51.100.7: icmp_seq=5 ttl=49 time=34.8 ms

--- 198.51.100.7 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 34.651/35.621/39.055/1.717 ms
```

Listing B.12: S4 - External IPv4 Ping

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-25 16:48 UTC
Nmap scan report for host-198-51-100-7.example.com (198.51.100.7)
Host is up, received user-set.

PORT STATE SERVICE REASON
23/tcp filtered telnet no-response
80/tcp filtered http no-response

Nmap done: 1 IP address (1 host up) scanned in 3.11 seconds
```

Listing B.13: S4 - External IPv4 Nmap Scan (NAT Filtered)